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Principles For the Design of Visual Progress Bars to Improve Motivation in Online Learning

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DAKOTA STATE UNIVERSITY

**PRINCIPLES FOR THE DESIGN OF VISUAL
PROGRESS BARS TO IMPROVE MOTIVATION IN
ONLINE LEARNING**

A doctoral dissertation submitted to Dakota State University in partial fulfillment of the
requirements for the degree of

Doctor of Philosophy

in

Information Systems

March 24, 2025

By

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DAKOTA STATE
UNIVERSITY.

DISSERTATION APPROVAL FORM

This dissertation is approved as a credible and independent investigation by a candidate for the Doctor of Philosophy degree and is acceptable for meeting the dissertation requirements for this degree. Acceptance of this dissertation does not imply that the conclusions reached by the candidate are necessarily the conclusions of the major department or university.

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“An investment in knowledge pays the best interest.” — Benjamin Franklin

ABSTRACT

Progress bars are used in many user interfaces to indicate user progress or wait time. Their main role is to set user expectations, and they can be auditory or visual. The use of visual progress bars is found in many applications, including games and is accepted as a strategy for gamifying user experiences. However, little research exists regarding the design of this tool, especially in how to leverage design elements to impact motivational affordance. Since progress bars are an element to gamify user interfaces, it benefits researchers and designers to have a deeper understanding of how to design the tool to increase motivation. Specifically, this research answers the following research question: What progress bar design principles improve motivational mechanics?

The research utilized the Design Science Research Methodology to create progress bar design principles based on elements related to motivational mechanics, motivational affordance, and user experience. Existing literature, focus groups, and user feedback were used to shape and refine these principles. The resultant artifact was evaluated through a review of a prototype incorporating the design principles using semi-structured interviews. The results are 11 design principles dealing with color, shape, text, placement, and animation of progress bars. The research provides design principles for progress bar design that concentrates on improving motivational mechanics in online adult learning environments.

Keywords: Gamification, Progress Bar, Design Principles, Motivation, Online Education

DECLARATION

I hereby certify that this dissertation constitutes my own product, that where the language of others is set forth, quotation marks so indicate, and that appropriate credit is given where I have used the language, ideas, expressions or writings of another.

I declare that the dissertation describes original work that has not previously been presented for the award of any other degree of any institution.

Signed,

A handwritten signature in black ink, reading "Tania Williams". The signature is written in a cursive style with a large initial 'T' and a long, sweeping underline.

Tania Williams

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CHAPTER 1

INTRODUCTION

Background of the problem

Progress bars are used as a means to track and display user progression or to display wait time and are intended to provide a user's status in relation to a specific goal (Nah et al., 2019). They are used to gamify online experiences, and their use has been shown to benefit self-directed learning (Zhu & Bonk, 2019). As a design element, they align with various principles of gamification including challenge, interactivity, competition, achievement, and reinforcement (Nah et al., 2019). Additionally, they have been shown as a tool to set user expectations regarding time in a task or how long a user can expect to wait (Harrison et al., 2007). Yet, currently, there has been only nominal research in how to design progress bars for tracking and displaying user progression in a way that improves user motivation in general, much less in the area of gamifying online learning. Gamifying online learning is an example of gamification, which is defined as using elements associated with game design in a non-game context (Deterding et al., 2011). The word *gamification* is claimed to have been coined in 2002 by Nick Pelling, who wished to add a game-like user interface with commercial electronic transactions (Pelling, 2011). However, researchers have been exploring how to use the features of computer games to make learning more interesting since the 1980s (Malone, 1980).

For this research, gamification coincides with the idea of gameful design -- utilizing game design elements to create gamefulness (Deterding et al., 2011). Specifically, the research is rooted in the design of an interaction design component often utilized in games but leveraged in a non-game context to provide a gameful experience (Deterding et al., 2011).

Statement of the problem

When looking at the current state of gamification research, studies often treat gamification as a uniform concept instead of focusing on specific gamification elements (Arce & Valdivia, 2020; Durin et al., 2019; Filippou et al., 2018; Oliveira et al., 2018; Westenhaver et al., 2022); however, as noted by Sailer et al. (2017), researchers should consider the impact of

distinct game design elements on the entire gamification environment as well. Unfortunately, this has not been the case. In fact, when making decisions as to which design elements should be included in environments, the choices are often arbitrary and subjective (Mazarakis & Bräuer, 2023; Sailer et al., 2017).

One element that is often included in gamified learning environments is a progress bar. Progress bars can serve one of two functions (Nah et al., 2019): 1. they can indicate user progression, as is their purpose in gamified learning environments, or 2. they can indicate wait time, as seen when downloading software or waiting for a site to load. The research on progress bar design to date has mainly focused on their function as an indicator of wait time (Bovermann et al., 2018).

When researchers have considered progress bars as a function to indicate user progression, only a limited amount of study of the tool has been done outside of research pertaining to user satisfaction or research in conjunction with other gamification elements, such as leaderboards or badges (Bovermann et al., 2018; Villar et al., 2013). Since, progress bars are often included in training modules or other user interactives, where progress bars serve as an indicator of user progress, understanding how to construct progress bars to fully serve their purpose in gamified learning is needed.

To date, researchers have touched on progress bars in relation to various training fields, such as those of healthcare (Comello et al., 2016; Westenhaver et al., 2022). However, again, the focus of the research has been on the impact of the existence of progress bars on user engagement. Very little research exists on the actual design of progress bars as they are used in training modules, specifically training modules in an adult online educational setting.

As efforts to gamify online training and educational resources increase, it is important for designers and researchers to better construct progress bars to maximize their ability to motivate users in adult online training modules. When considering motivation it can be viewed as the catalyst and direction of behavior, where motivation is the meeting of needs and direction is the process and structures that direct action toward the meeting of needs (Deci & Ryan, 1985). Therefore, motivation is the process that starts and sustains activity toward a goal (Schunk & DiBenedetto, 2020). Motivation can be divided into two states: entering motivation, which is deciding to participate, and task motivation, which is the effort required to stay on task and persist (Garrison, 1997). When considering the role of a progress bar in adult online training

modules, the tool better aligns with task motivation, as its function relates to persisting towards a goal that has already begun. The motivational variables tied to task motivation are “perceived task value, task enjoyment, perceived task difficulty, and degree of effort” (Guo et al., 2023). Even trivial-task competition (completion of sub-tasks) enhances a participant’s motivation to complete larger tasks (Guo et al., 2023). This means that perceived user progress, such as progress demonstrated through a progress bar, should motivate users. Therefore, this paper provides research regarding the design of visual progress bars as they are used in adult online training modules. The research aims to answer the following research question:

What progress bar design principles improve motivational mechanics?

Objectives of the dissertation

The research fills a gap related to the intentional design of progress bars, especially in the areas of education and training. To date, researchers have focused on progress bars as a design element, with very few examining the design of progress bars independently of other gamification elements. When looking at the use of progress bars independently, the focus has been on the impact of their inclusion on user engagement, without factoring in the bar’s design. Engagement; which focuses on active participation (behavioral engagement), reactions (emotional engagement), and psychological investment (cognitive engagement); is tied to constructs such as effort, attention, persistence, enjoyment, and goal setting (Karimah & Hasegawa, 2022). However, motivation, as described previously, starts and sustains an activity toward a goal. In a sense, engagement provides a purpose, while motivation drives the user to fulfill the purpose (Martin et al., 2017).

Research related to progress bar design has been limited to progress bars as a display of wait time, with research as to how to better design them to increase motivation in educational-related environments not being explored. As such is the case, this research into the creation of design principles for visual progress bars to positively impact user motivation in educational-related online environments allows designers to customize the design element to better suit the target audience and enhance the impact of the tool. Current research ties progress bars to gamification principles or strategies such as challenge, interactivity, competition, achievement, and reinforcement, meaning that a better design of progress bars will provide more insight into how to maximize the design of the tool to maximize user experience (UX). In order to develop

visual progress bar design principles to support user motivation in online learning environments, the researchers utilize extant literature, focus groups from experts in the field, and user feedback.

CHAPTER 2

LITERATURE REVIEW

This chapter discusses the literature on progress bars in gamification and general progress bar design to better understand current work in progress bar design in online learning environments. It summarizes the literature and concludes with the gap in existing research.

Role of progress bars in gamification

As environments are sometimes gamified to foster motivation to improve learning (Sailer et al., 2013), researchers have attempted to identify gaming elements that lend to gamifying these environments. These lists of elements vary from researcher to researcher. However, progress bars are usually included in these design elements, along with other elements such as points, badges, leaderboards, progress bars, quests, meaningful stories, and avatars (Sailer et al., 2013). Nah et al. (2019), compiled a much longer list, including other elements such as levels, rules, and use of teams; additionally, the researchers aligned these design elements with eight basic principles or strategies for gamification, with progress bars to being tied to five of these principles (challenge, interactivity, competition, achievement, and reinforcement).

Unfortunately, when examining the gamification of online learning environments, researchers tend to apply all the game design elements at once without examining their individual effects (Mazarakis & Bräuer, 2023). However, some researchers have provided descriptions of the intended role of the progress bar in the gamified experience, and in some cases, progress bars are associated with multiple principles within the same study. Table 1 provides examples of how the role of progress bars is defined in research.

Table 1. *Role of progress bars as described in research*

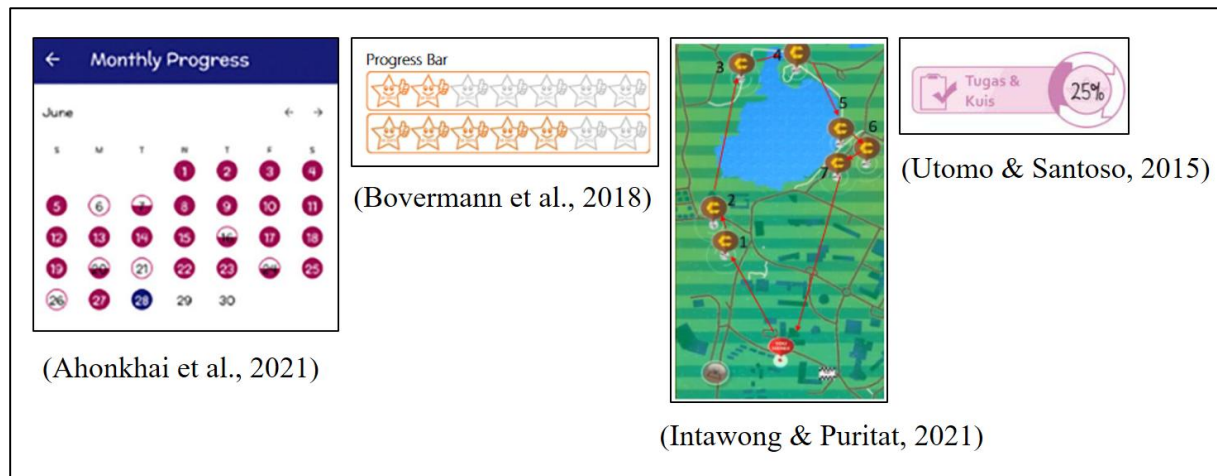
Role	Definition	Authors
Listed as a design element	No role provided	(Arce and Valdivia 2020; Filippou et al. 2018; Oliveira et al. 2018)
Challenge	Learning growth and development and opportunity for advancement	(Bovermann et al., 2018; Molenaar et al., 2019; Nah et al., 2019; Ninaus et al., 2015; Sailer et al., 2013; Utomo & Santoso, 2015; Zhu & Bonk, 2019, 2020)
Interactivity	Immediate feedback	(Bedwell et al., 2022; Intawong & Puritat, 2021; Meşe & Dursun, 2018; Nah et al., 2019; Ninaus et al., 2015; Sailer et al., 2013; Utomo & Santoso, 2015)
Competition	Social comparisons	(Nah et al., 2019)
Achievement	Recognition of accomplishment	(Ahonkhai et al., 2021; Bovermann et al., 2018; Intawong & Puritat, 2021; Nah et al., 2019; Ninaus et al., 2015)
Reinforcement	Structure of rewards based on performance	(Bovermann et al., 2018; Coppola et al., 2024; Mazarakis & Bräuer, 2023; Nah et al., 2019; Ninaus et al., 2015)

These roles of progress bars partially align with how progress bars are used in other user interfaces. For example, researchers found that using progress bars to denote time for message delivery reduced stress and mental workload by providing immediate feedback to the users (Schor, 2024). Conrad et al. (2010) found, with online surveys, progress bars can impact perceived task difficulty and duration in a task -- especially with progress information shown early in the survey -- again showing a link between progress bars and interactivity.

Progress bar design

When looking at existing research that mentions the design of progress bars, the bars can be auditory, depending on tones or voices to indicate progress (Kortum et al., 2011). Of the visual progress bars described, some researchers have examined the design of a functional progress bar, which is the traditional rectangle bar (Molenaar et al., 2019; Ninaus et al., 2015). Others have used hedonic progress bars, meaning the bar was a shape other than a rectangle (Ahonkhai et al., 2021; Bovermann et al., 2018; Intawong & Puritat, 2021; Utomo & Santoso, 2015) (see Figure 1). Examples of these designs include a map with a route that is filled in as the user moves, a calendar-shaped map with circles that are filled in for each day, a row of stars, and

circles to indicate competition. Researchers have also focused specifically in the success of hedonic versus functional progress bars related to wait time (Lee et al., 2017).



Note. From “PEERNaija: A Gamified mHealth Behavioral Intervention to Improve Adherence to Antiretroviral Treatment Among Adolescents and Young Adults in Nigeria,” by Ahonkhai et al., 2021, *Frontiers in Reproductive Health*, 3, p. 9. From “Online learning readiness and attitudes towards gaming in gamified online learning – a mixed methods case study,” by Bovermann et al., 2018, *International Journal of Educational Technology in Higher Education*, 15, p. 16. From “A Framework of Developing Mobile Gamification to Improve User Engagement of Physical Activity: A Case Study of Location-Based Augmented Reality Mobile Game for Promoting Physical Health,” by K. Intowong and K. Puritat, 2021, *International Journal of Online and Biomedical Engineering (iJOE)*, 17(7), p. 110. From “Development of gamification-enriched pedagogical agent for e-Learning system based on community of inquiry,” by A. Utomo and H. Santoso, 2015, *Proceedings of the International HCI and UX Conference in Indonesia*, p. 6.

Figure 1: Examples of Hedonic Progress Bars from Literature

Outside of gamification efforts, Molenaar et al. (2019) explored progress bars through the context of survey/goal attainment, looking at progress bars as a strategy for monitoring task completion. There are a few examples of research that focus on the design of progress bars themselves. Kortum et al. (2011) and Lee et al. (2017) examined progress bars in relation to wait time, with one dealing with customers’ use of auditory progress bars tied to a phone system and the other dealing with the use of functional and hedonic progress bars through the use of a travel website.

Research gap

The existing research lacks an examination of the intentional design of progress bars, particularly related to how to design visual progress bars in contexts where their design is to afford user motivation. While researchers have focused on progress bars as a design element, very few have examined the design of progress bars independently of other gamification

elements. These efforts have been related to wait time and have yet to be regarding user progress. Some researchers have considered the use of progress bars independently, but the focus has been on the impact of the inclusion of a progress bar on user engagement without considering the bar's design.

Focused research on progress bar design has been limited to progress bars as a display of wait time. As a result, research as to how to better design a progress bar to increase motivation in educational-related environments has yet to be explored. This research into design principles for visual progress bars to impact user motivation in educational-related online environments provides a conceptual model for progress bar design. The principles provide guidance on how to better suit the target audience and enhance the tool's impact. A conceptual model offers a “set of constructs and rules that show how to combine the constructs to model real-world domains,” supporting an understanding of the relationship among entities, in this case the understanding among various factors in progress bar design with a resulting progress bar to positively impact motivation (Wand et al., 2002).

CHAPTER 3

THEORETICAL FOUNDATION

This chapter presents a discussion on the theoretical background of this research, including the relationship among motivation, motivational affordance, and user experience (UX). It also provides the research question that guided the study.

As the focus of the research is to develop design principles for progress bar design to increase user motivation in an adult educational module, it is important to explore the role of progress bars in motivation.

When looking at andragogic strategies for increasing motivation in an online environment, one strategy is gamification (Włodkowski, 2017). As progress bars are an element of gamification, the inclusion of such a design element is a valid approach to increasing motivation. The addition of the element in adult online learning is also supported by its being classified as a feedback tool, since another strategy for motivation for the adult learning in an online learning environment is providing effective feedback (Włodkowski, 2017). As progress bars are a tool for providing feedback regarding progress within an online environment, the addition of this type of design element should assist in motivating users.

As far as focusing specifically on user motivation, prior research has tied progress bars to motivational mechanics since progress bars provide clear and achievable goals and direct feedback (Sailer et al., 2013). Sailer et al. (2013) break down motivational mechanics further to define it through six principle perspectives (trait perspective, the behaviorist learning perspective, the cognitive perspective, the perspective of self-determination, the perspective of interest, and the perspective of emotion), with each perspective providing an implication for practice. They tie progress bars to two of these perspectives. The first perspective is the cognitive perspective, with progress bars being seen as a tool for means-ends analyses -- addressing user goals, expectations and values of consequences by providing a clear and achievable goal. The second perspective they link to progress bars is the perspective of interest. As Bose (2008) notes, *“Information designers need to be aware of how users relate themselves within a particular construct; this is the primary connection between flow and information design.”* This feeling of flow, explored by Csikszentmihalyi et al. (2005), is a current that carries participants effortlessly

through a task. When experiencing flow, participants forget “*time, fatigue, and everything else but the activity itself*” (Csikszentmihalyi et al., 2005). The characteristics of flow include merging of action and awareness, a sense of control, and an altered sense of time (Csikszentmihalyi et al., 2005). In this case, progress bars enhance the feeling of flow by providing direct feedback and a clear goal in the context of contributing to interest in the task or environment. Flow is achieved by providing a clear set of goals, creating a balance between perceived challenges and perceived skills, and providing clear immediate feedback (Csikszentmihalyi et al., 2005).

Several researchers have linked flow with motivation. For example, Zairi et al. (2022) found that students who were motivated to play serious games experienced a higher level of flow, noting the reciprocal relationship between the two constructs. This is supported by Garris et al. (2002) who maintained that feedback, by giving users an understanding of progress toward a goal, motivates the user to provide a higher degree of effort and attention on a task. Therefore, a better progress bar design can better drive user motivation by considering a design’s ability to provide a clear and achievable goal and direct feedback to the user in an interesting way.

As discussed previously, progress bars can have auditory or visual components. In existing research (Kortum et al., 2011), auditory progress bars closely align with wait-time by providing information as to how long a person must wait for a service. However, in the context of online learning, the purpose of the element is to provide an indicator of movement towards a goal and the use of auditory elements could interfere with other components, such as character dialogue, content narration, or music. Therefore, the development will center around design principles tied to visual progress bars.

When considering elements of visual progress bar design, it is valuable to consider motivational affordance and UX, as just the inclusion of a design element typically tied to gamification does not necessarily guarantee a motivated user (Deterding, 2011). In fact, Deterding, (2011) notes there is a lack of theoretical models that link “to the more granular level of single interface of game design patterns” (p. 2). Therefore, when focusing on progress bar design, it is not only important to take into account elements of task motivation, such as task enjoyment, it is also necessary to consider how the features of an object support the outcome of motivation to provide user satisfaction (Deterding, 2011). This implementation of motivational properties is called motivational affordance (Berglund et al., 2022). Taking these considerations

into account, the relationship among the constructs tied to game element design can be seen in the following concept map (see Figure 2).



Figure 2: *Relationship Among Motivation, Motivational Affordance, and User Experience*

The concept map shows the relationship of constructs dealing with the design of a single gamification element, with motivation being a desired outcome, motivational affordance being the implementation of the artifact to achieve the outcome, and UX being the evidence of the outcome -- in this case, a motivated user. This is similar to the three components of gamification proposed by Berglund et al. (2022), with the authors also seeing motivation feeding into the implementation of motivational affordances; however, they tied these constructs to psychological and behavioral outcomes in relation to gamification in general. They did not attempt to explore the relationship between these constructs in relation to the design of a specific gamification element, nor did they consider the more nuanced aspects of motivational affordance, as demonstrated later in this section.

Building on this relation among motivation, motivational affordance, and UX, it is crucial to understand the role of specific gamification design elements in relation to these constructs. However, before adding these gamification design elements, it is important to account for the more nuanced characteristics of motivational affordance. According to Deterding (2011), these motivational affordances can be broken down into two categories: situational motivational affordances, which deal with the motivation of the user tied to the context of the play (situation), and artifactual affordances, which harken back to the design and appeal of the artifact itself (see Figure 3).

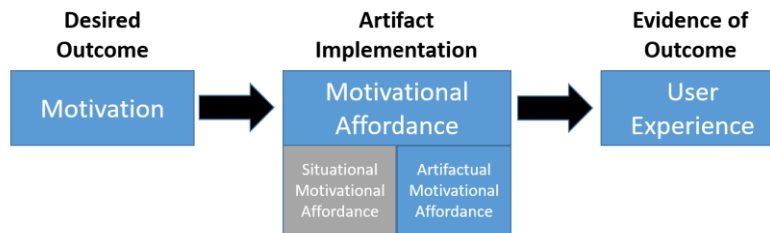


Figure 3: *Relationship of Motivational Affordance Types in Progress Bar Design*

As situational motivational affordance is dependent on the abilities or characteristics of the user, it is not a consideration to the design of progress bars, but it is included in Figure 3 to reflect its role in relation to motivational affordance. As artifactual motivational affordance is tied to the artifact's design and appeal, it is an area impacted by artifact design. This is shown by Figure 4, which demonstrates how artifact design is a vehicle to achieve artifactual motivational affordance.

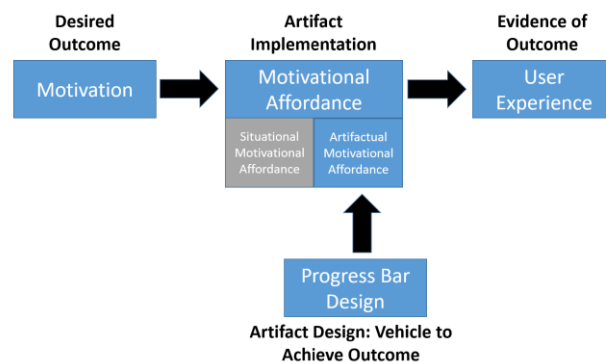


Figure 4: *Role of Progress Bar Design in Achieving User Motivation*

After establishing the role of progress bar design in the context of user motivation, the next task is to tie in the specific design elements of visual progress bars. When looking at how data can be visualized, we can take into account the factors of representation, presentation, and interaction (Spence, 2014) (see Figure 5).

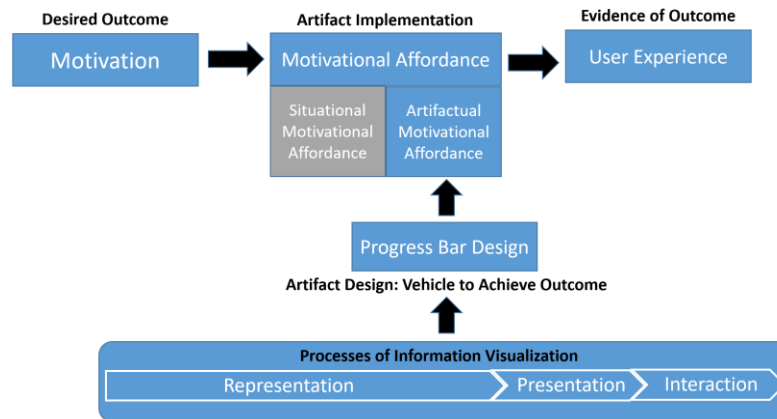


Figure 5: Relationship Among Processes of Information Visualization and Motivational Affordance

In this case, *representation* is defined as the selection of techniques to illustrate important concepts (object color, shape, and text); *presentation* is defined as the space, time, and human visual system (object placement); and *interaction* is defined as the coupling of input and output (object animation) (Spence, n.d.; Ware, 2004). Therefore, color, shape, text, placement, and animation are included in Figure 6.

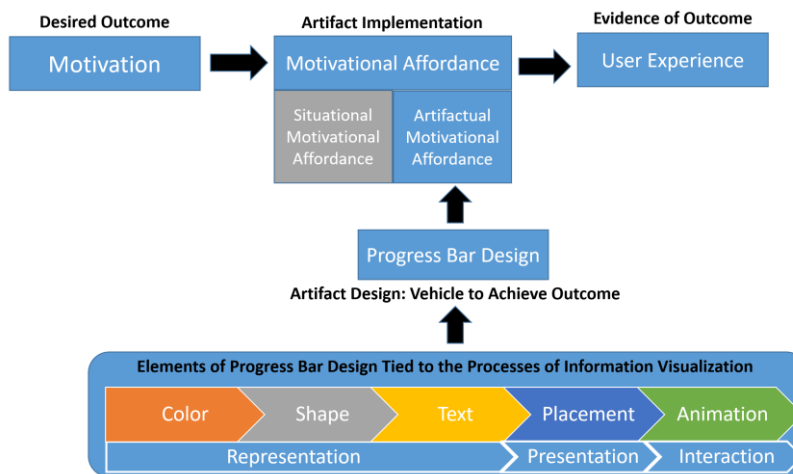
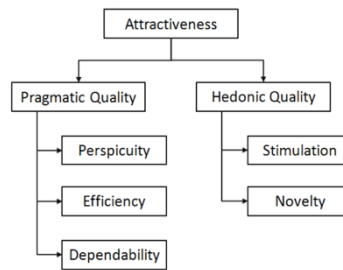


Figure 6: Role of Progress Bar Design Elements in Achieving User Motivation

Upon understanding the role of progress bar design elements in user motivation, the next step is to conceptualize how user motivation is exhibited. To promote artifactual motivational affordance, the elements of visual progress bar design must also support UX. Aspects of UX include attractiveness, perspicuity, efficiency, dependability, stimulation, and novelty (Schrepp et al., 2017). Attractiveness is an emotional reaction, while perspicuity, efficiency, and dependability relate to tasks or goals the user aims to reach (pragmatic quality). Stimulation and novelty are associated with pleasure or fun (hedonic quality). When considering the relationship

among these six aspects of (UX), Schrepp et al. (2017) state that pragmatic quality and hedonic quality influence attractiveness (see Figure 7).



Note. From “Design and Evaluation of a Short Version of the User Experience Questionnaire (UEQ-S),” by M. Schrepp, A. Hinderks, and J. Thomaschewski, 2017, *International Journal of Interactive Multimedia and Artificial Intelligence*, 4(6), p. 104.

Figure 7: Assumed Scale Structure of the User Experience Questionnaire (UEQ)

These aspects of UX can be mapped to concepts tied to user motivational mechanics related to progress bars. The pragmatic quality of UX aligns with the cognitive perspective of motivational mechanics, as both are concerned with meeting the user’s goals and expectations. The hedonic quality of UX aligns with the perspective of interest in motivational mechanics, as both are tied to the user’s immersion in the environment. Attractiveness is included in both areas, as attractiveness is seen as being influenced by both qualities (Sailer et al., 2013; Schrepp et al., 2017). Table 2 demonstrates this alignment between motivational mechanics and aspects of UX.

Table 2. Alignment Between Motivational Mechanics and User Experience

Quality	Motivational Mechanics Tied to Progress Bars (Sailer et al., 2013)	UEQ Aspects (Schrepp et al., 2017)
User’s goals and expectations	Cognitive perspective	Pragmatic: Perspicuity, Efficiency, Dependability, Attractiveness
User’s immersion in the environment	Perspective of interest	Hedonic: Stimulation, Novelty, Attractiveness

This alignment among the characteristics of a positive UX and the outcome of motivating the user should be considered in visual progress bar design. The opportunities of visual progress bar design to impact motivational mechanics and UX are reflected in Figure 8 shown below. The common design elements of color, shape, text (if used), placement, and animation have potential to increase motivational affordance as evidenced by an improved UX. This relationship was used to develop instantiations of progress bars for the research to inform in design principle creation.

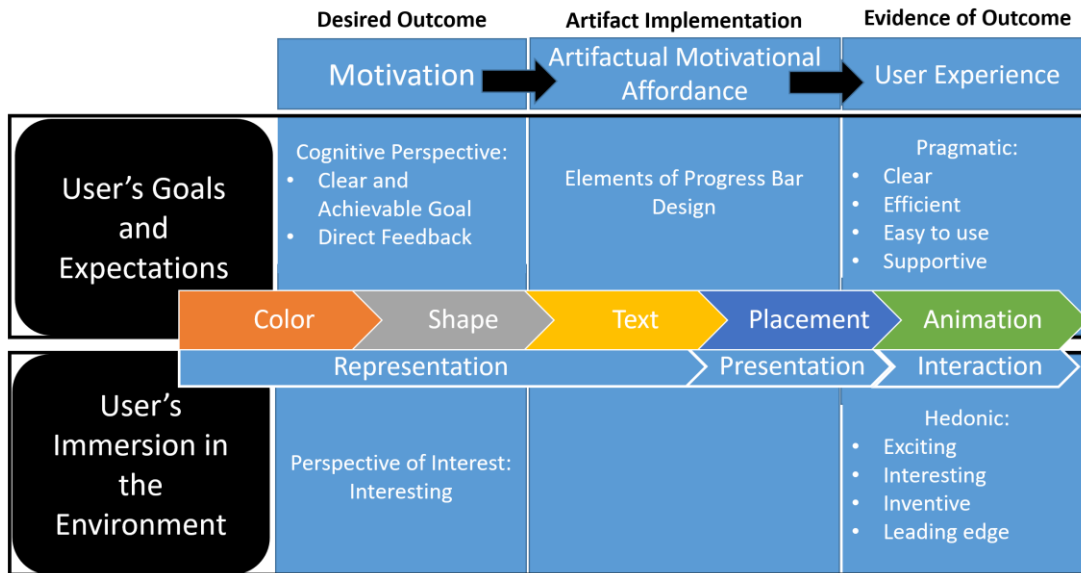


Figure 8: Opportunities to Impact Motivation through Progress Bar Design

CHAPTER 4

RESEARCH METHODOLOGY

This chapter provides the research methodology concerning the design of the artifact and outlines the process for evaluating the artifact. As noted by Tengler et al. (2021), the development of design principles should be iterative and may include the development of preliminary design principles and preliminary artifacts to shape the principles. Similar to their technique, the researchers chose to utilize a prototyping cycle, producing progress bar prototypes, conducting interviews with experts in the field to receive feedback on the prototypes to determine best practices for progress bar design, refining a selected prototype for user feedback, and utilizing the user feedback to guide further design principles development (see Figure 9). This process is also supported by Lockl et al. (2020); those researchers derived initial objectives from streams of literature, integrated those into a prototype, demonstrated the prototype, and used the feedback to refine the principles through an iterative approach.

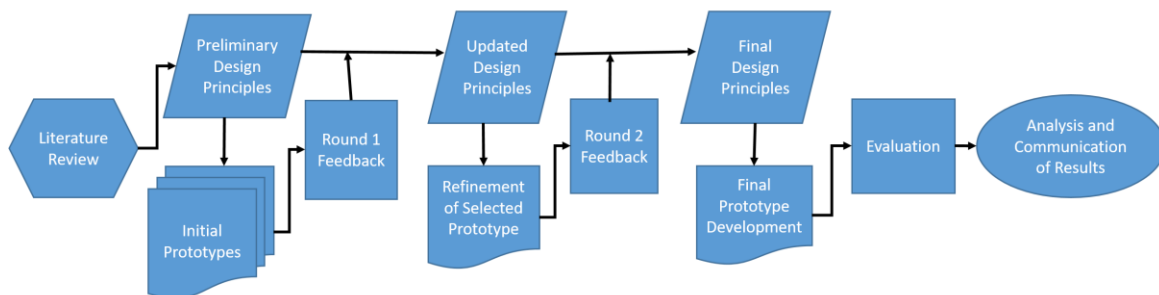


Figure 9: Design Principle Development Process

This process aligns with the design science research methodology (DSRM) outlined by Peffers et al. (2007). DSRM breaks design science research into six steps, with four possible research entry points. The six steps are identifying the problem and motivation, defining the objectives of a solution, designing and development, demonstrating the solution, evaluating, and communicating (Peffers et al., 2007). This outline is shown in Table 3 below.

Table 3. *Approach for Progress Bar Design Principles Development*

Problem Identification and Motivation	Solution Objectives	Design and Development	Demonstration	Evaluation	Communication
Literature review, preliminary design principles and initial prototype creation	Literature review	Focus groups, update of design principles, refinement of selected prototype, user feedback and analysis, and further refinement of the design principles	Integration of refined design principles into the final prototype	Focus group evaluation of the prototype	Presentation and publishing of the results

Identify problem and motivate

Problem identification and research motivation were accomplished through a literature review and identification of research gaps (see Chapter 2). The review revealed a gap in research related to the intentional design of progress bars, especially in the areas of education and training. Therefore, this paper addresses the following research question: What are progress bar design principles to improve motivation?

Define the objective for a solution

After identifying the research gap, the literature was also consulted in order to define the objective for a solution. The following was the design objective:

- To develop design principles that best support user motivation and provide a positive user experience.

Building on the alignment between motivational mechanics and UX (see Table 2), the objective was to provide progress bar design principles to support both the cognitive perspective and the perspective of interest. Thereby, the progress bar design principles should inform on creating a positive UX by providing perspicuity, efficiency, dependability, stimulation, novelty, and attractiveness.

Chapter 3 covers the relationship between motivation and UX, how motivational elements contribute to a user's goals and expectations and the user's immersion in the environment, and how the results of this feeds into the overall UX.

Design and development

The indicators for a positive UX were used as a starting point to determine an early draft of the design principles. The researchers utilized these, along with other guidance from literature, to develop preliminary design principles, which focused on the qualities of a progress bar, including color, shape, text (if used), placement, and animation. Using these preliminary design principles, the researchers developed prototypes, which were provided to experts for focus group review (Round 1 feedback) using the user experience questionnaire (UEQ). UEQ was originally developed in Germany by Laugwitz et al. and was later translated and tested in English (Laugwitz et al., 2008). It was created using a data analytics approach to measure distinct aspects of UX. These aspects include attractiveness, perspicuity, efficiency, dependability, stimulation, and novelty. The original tool had 26 items, which randomized the polarity of the items (order of the positive or negative term tied to the aspects) (Laugwitz et al., 2008). In order to shorten the UEQ, Schrepp et al. (2017) designed and evaluated a shorter version of UEQ, UEQ-S (Figure 10), to be used in specific situations, including a situation where “a participant is asked to judge the user experience of several products or variants of a product in one session.” Participants were asked to review multiple variations of progress bars using the UEQ-S.

obstructive	o o o o o o o	supportive	1
complicated	o o o o o o o	easy	2
inefficient	o o o o o o o	efficient	3
clear	o o o o o o o	confusing	4
boring	o o o o o o o	exciting	5
not interesting	o o o o o o o	interesting	6
conventional	o o o o o o o	inventive	7
usual	o o o o o o o	leading edge	8

Note. From “Design and Evaluation of a Short Version of the User Experience Questionnaire (UEQ-S),” by M. Schrepp, A. Hinderks, and J. Thomaschewski, 2017, *International Journal of Interactive Multimedia and Artificial Intelligence*, 4(6), p. 105.

Figure 10: Short Version of UEQ-S provided by Schrepp et al. (2017)

During round one, two focus group sessions were held, each using the UEQ-S. The focus group participants were a group of nine experts with backgrounds as curriculum developers, educators, and UX designers. The participants provided suggestions for improving the prototypes. These suggestions guided principle design. Each focus group session lasted about one hour. About 10 minutes before the end of the session, the researcher began wrapping up the discussion and alerted the group to the amount of time left. The focus group session was recorded, with participants being notified of the plans to record ahead of time (Litosseliti, 2003).

The survey results and the focus group discussion were analyzed, with the instantiation with the highest UEQ-S results being selected to inform on future design principle design efforts. Member checking was used to provide the focus group members the opportunity to provide feedback on the validity of the findings (Creswell, 2012, p. 259).

The feedback from Round 1 is described in Chapter 5 and was used to select one of the prototypes for further principle development. The focus group provided suggestions for improving the chosen prototype, which were noted for further principle design development and were implemented into a prototype to be placed into the online adult learning environment. The progress bar design prototype with the best UEQ-S results was improved using updated design principles and integrated into an online adult learning environment for Round 2 feedback.

Once the selected prototype was updated based on the refined principles, users in Round 2 compared the prototype to a traditional progress bar to determine the progress bars' ability to increase user motivation in an online learning environment, specifically, the progress bars' ability to sustain activity towards completion of the learning module. This information was used to identify gaps in the draft of the design principles. This application of applying the principles to a prototype in order to advance design principle development is similar to other design principle design approaches (Hulshof, 2010; Marx et al., 2011; Tengler et al., 2021).

For Round 2, the researchers used user surveys from a population of college students for data collection for this comparison. Amazon gift cards were provided to participants for participating in the study, and the study was limited to up to 50 participants, with 42 students recruited and 41 students fully participating in the study. Students were recruited from cybersecurity and communication courses and through various student clubs. Semi-structured interviews were used to collect feedback. This strategy is in keeping with Gable (1994), who noted that the format could be used "to develop contextual richness that is valuable in model building."

The data collection questions utilized the same UEQ-S survey for both the traditional progress bar and the progress bar version refined using focus group feedback. Additionally, the participants were asked questions to justify their answers. To understand how perceived motivation increased, participants were asked if the progress bar design increased motivation. If the participant indicated that it did increase motivation, the participant was asked about those features of the progress bar promoting motivation. If the user indicated the progress bar

undermined motivation, the participant was asked what about the progress bar undermined motivation. ATLAS.ti was used to code and analyze the open-ended responses, using a descriptive framework that aligns with the data collection instrument (Yin, 2014, p. 140). As for an analytical strategy, the researchers used pattern matching (Yin, 2014, p. 147) and coding (Strauss & Corbin, 1998). The researchers used preliminary coding to apply filters to better understand coding trends. Then the researchers next examined the data multiple times for themes related to progress bar design (Patton, 2002).

These responses were used to refine the design principles. Participant responses were analyzed and coded to understand reoccurring themes related to the design of progress bars. These were used to further define the resultant design principles.

Demonstration

Next, these final design principles were integrated into a prototype for final evaluation. This integration was to test the feasibility of applying the principles to a specific problem (Peffer et al., 2007). Using the refined design principles, the researchers made the following changes:

- Adjusted the colors of the progress bar.
- Adjusted the size, shape, and level of detail of the progress bar.
- Included more frequent updating of the progress bar.
- Segmented the progress bar to better match the amount of content remaining.

Colors were adjusted to make parts of the progress bar more visible and to better complement and connect with the rest of the content. This included colorizing areas that had previously been gray to incorporate more of the colors from the rest of the learning module.

The progress bar was adjusted to better scale across various display formats. This adjustment involved making some components larger to make them easier to see, adjusting the shape of some elements to allow for better segmentation of the bar, and more milestones to reflect the more accurate segmentation. Segmentation changes were made to allow for more frequent updating of the progress bar and to better reflect the amount of content remaining in the learning module.

The prototype was integrated into a learning module for user evaluation.

Evaluation

Following the implementation of the design principles into a prototype, the researchers invited users to review the created prototype integrated into a learning module to understand how the prototype incorporating the design principles compared to a default progress bar in relation to factors pertaining to UX. This approach is similar to the interview format used in design principle development evaluations by Sturm & Sunyaev (2019), where the researchers asked a series of open-ended questions about a prototype to get feedback on the implemented design principles.

Once the prototype was adjusted to reflect the design principle refinement, the prototype was used to assess the understandability and usefulness of the design principles to inform progress bar designers and developers on how to design progress bars to improve motivation. The researchers invited users to review the created prototype, which had been integrated into a learning module. This approach allowed researchers to understand how the instantiated design principles in the form of a prototype compared to a default progress bar in relation to factors pertaining to UX.

This approach is similar to the interview format used in design principle development evaluations Sturm & Sunyaev (2019), where the researchers asked a series of open-ended questions about a prototype through semi-structured interviews using a focus group format.

Subjects

Subjects included undergraduate and graduate students enrolled in a course at a university. They were recruited through email solicitation and social media postings. They were a population of convenience, with the researchers extending the invitation for participation to students across the whole university. The researchers sought interviews from a minimum of nine participants, which is consistent with other research of this nature (Sturm & Sunyaev, 2019). However, sessions were analyzed and repeated until theoretical saturation was reached and no new or relevant data seemed to emerge (Strauss & Corbin, 1998).

The study was limited to determining the progress bar design principles' ability to motivate users compared to a traditional progress bar. This comparison evaluation of a new artifact with an existing artifact was used to determine if the implemented design principles result in an improvement of motivation (Venable et al., 2016).

Instruments of measurement

The researchers utilized the same tools used in with the focus group participants earlier in the research (rounds 1 and 2), which was the condensed version of the UEQ, along with digital representations of the prototype instantiating the design principles and a traditional progress bar. Both representations were placed inside of the identical learning modules, with the progress bar being the only difference between the learning modules.

The researchers used semi-structured interviews for data collection. These consisted of questions regarding the aspects of UX (attractiveness, perspicuity, efficiency, dependability, stimulation, and novelty) and revolved around the short version of the UEQ (Schrepp et al., 2017).

Interviews

Eleven people responded to the recruitment materials, with all being invited to participate in the study. Of those, all eleven were interviewed. The interviews were recorded, and participants were notified of the intent to record prior to the interview.

Participants were asked to review both versions of the learning module, with Version A being the traditional progress bar and Version B being the prototype incorporating the design principles.

The researchers used the condensed version of the UEQ, as a guide for participant review. Adhering to a focus-group interview format, the meetings began with introductions and then researchers asked questions regarding the aspects of UX (attractiveness, perspicuity, efficiency, dependability, stimulation, and novelty) (Schrepp et al., 2017). Participants were invited to explain or discuss answers through group discussion during semi-structured interviews (see Appendix A). The session lasted about one hour, with the researchers inviting participants to provide closing thoughts and comments 10 minutes before the end of the session. Participants were given the opportunity to review the feedback on the validity of the findings as a form of member checking (Creswell, 2012, p. 259).

Data Analysis

ATLAS.ti was used to code and analyze the interview responses. As part of the analytical strategy, the researchers used pattern matching (Yin, 2014, p. 147) and open and selective coding

(Strauss & Corbin, 1998). The findings associated with this evaluation can be found in the Results chapter.

CHAPTER 5

PRINCIPLE DEVELOPMENT

To develop progress bar design principles to support progress bar design to improve user motivation, the researchers relied on existing literature and focus group and user feedback. This development is described in this chapter.

Preliminary design principle development

The design objective was to identify design principles that best support user motivation and provide a positive UX. The researchers drew from existing literature concerning the alignment between motivational mechanics and UX (see Table 2); the objective was to provide guidance on progress bar design principles to support both the cognitive perspective and the perspective of interest. The researchers chose to use positive UX as guidance for measuring the support of user motivation and as a starting point for the early drafts of the design principles. Chapter 3 covers the relationship between motivation and UX, how motivational elements contribute to a user's goals and expectations and the user's immersion in the environment, and how the results of this feeds into the overall UX.

Guidance from literature was also utilized, specifically research related to the characteristics of progress bars, including color, shape, text (if used), placement, and animation, through the lens of representation, presentation, and interaction (Spence, 2014). The research also took into account factors such as color psychology, cognitive cueing, and accessibility guidelines.

Color

Although color meaning can vary by user demographics, certain colors are often associated with particular symbols or emotions (Tulsiani, 2024). For instance, red is associated with urgency, while blue is associated with calmness or focus (Tulsiani, 2024). Therefore, these associations need to be considered in progress bar design.

Additionally, research regarding user preference was also considered. When considering color choice for cognitive cueing, it is important to factor in cognitive load. Cognitive cueing

refers to the use of visuospatial characteristics of instructional materials to assist learners in focusing attention and in organizing and integrating information (de Koning et al., 2009). Educational interfaces, often rich in display content to support these cognitive cues, do offer a potential for cognitive overload, which decreases the effectiveness of the tools (Roberts, 2008). Therefore, factors such as artifact color and placement should be considered to help alleviate these concerns (Roberts, 2008). When dealing with the function of progress bars as a cognitive cue, their role aligns best with the function of emphasizing the organization of instruction (de Koning et al., 2009). This means that the color of the progress bar should be distinct enough for identification, yet complimentary to the rest of the components (Roberts, 2008). While vivid and bright colors were tied to excitement, researchers found hot colors and jarring color schemes can contribute to cognitive overload; therefore, combinations such as blue/orange and violet/yellow were avoided (Deubel, 2003; Tulsiani, 2024). Researchers found neutral colors a preferred choice for adults, with blue being a color often testing well (AL-Ayash et al., 2015; Deubel, 2003; Osman & Hwang, 2021; Tulsiani, 2024).

Added to the concerns about color psychology and color preference is the need for color readability (Richardson et al., 2014). Certain color combinations, such as red/green or blue/yellow, can be challenging for color-blind users (Tulsiani, 2024). Low vision users also need high contrast in colors, and color alone should not be the only way to provide information, as doing so also creates visibility concerns (*Guidance on Web Accessibility and the ADA*, 2024). Hence, a color contrast checker could be used to ensure a contrast ratio of at least 3:1, which is recommended for graphics and user interface components (*WebAIM: Contrast Checker*, n.d.).

Therefore, two preliminary design principles were identified related to color: 1). Progress bar color should complement the rest of the content but should be distinct and recognizable, and 2). Progress bar designers should avoid jarring color schemes and ensure colors meet the needs of users with color blindness or low vision.

Shape

When considering visualizations related to learning materials, an object's appearance should capture attention (de Koning et al., 2009). However, designers should be cautious in adding objects, as learners could become over stimulated (Roberts, 2008). In fact, if a visual is determined to be unhelpful, it can be viewed as an obstacle (Nielsen & Pernice, 2010). However,

images that are related to the rest of the content have been shown to receive more attention by users and are likely to get the attention of users even if the image is of poor quality (Nielsen & Pernice, 2010). Nielsen & Pernice (2010) found, when tracking how people view elements of websites, that participants ignore or barely look at content they view as not related to the main idea of the page. In the case of progress bars, researchers considering the design of progress bars related to wait time found that an airplane-shaped progress bar on a travel website resulted in increased perceived enjoyment by users (Lee et al., 2017). Therefore, the following design principle was identified related to shape: Progress bar design should relate to the rest of the content.

Text

Another element related to progress bar design is the use the text. While text is not always used in the visuals, when it is, researchers and designers should consider the readability of the text. With this in mind, mixed case (mix of upper and lower case letters) was shown in previous research to be preferred by users (Osman & Hwang, 2021). This allows for the design principle: Progress bar text, if used, should have a mix of uppercase and lowercase text.

Placement

Progress bars also should be appropriately placed on the screen. Since it is important that related information be spatially grouped together, the placement of the progress bar should not break up related information (Vekiri, 2002). There is less cognitive load when images are related to the subjects of the page (Harper et al., 2009). The relevance of this is important as dynamic elements have the potential to override attention and ill-defined or out-of-context elements can have negative effects on learning (de Koning et al., 2009).

This gave rise to the design principle: Progress bar placement should complement the rest of the content.

Animation

Alignment to gamification principles also drove the design principles. Prior researchers have viewed progress bars as a design element that aligns to various principles of gamification and aligned it to five of the eight basic principles of gamification: challenge, interactivity, competition, achievement, and reinforcement (Nah et al., 2019), with the remaining three being

goal orientation, social connectivity, and fun orientation. They view these eight elements as a way to increase motivation, engagement, and performance. Therefore, by providing progress bar design principles to account for this, the researchers considered ways to motivate users by capturing principles to guide the design of “an innovative, purposeful artifact for a specified problem domain” (Hevner et al., 2004). As such, the design centered around supporting challenge, interactivity, competition, achievement, and reinforcement.

To ensure the design principles centered around supporting challenge, interactivity, competition, achievement, and reinforcement; the researchers considered how progress bars show movement and progression. Traditional progress bars often fill-in with a color as a user progresses. To account for this, the researchers added another design principle: 1). Progress bars should update to reflect user progression. The researchers added movement and/or colorization to the prototypes to support these gamification elements.

To reflect this design principle in the prototypes, the researchers added animation to the artifacts. Each prototype had phases to indicate user movement through a module. Table 4 provides an overview of the preliminary design principles.

Table 4. *Preliminary Design Principles*

Element	Characteristic	Preliminary Design Principle
Representation	Color	Progress bar color should complement the rest of the content but should be distinct and recognizable.
	Color	Progress bar designers should avoid jarring color schemes and ensure colors meet the needs of users with color blindness or low vision.
	Shape	Progress bar design should relate to the rest of the content.
	Text	Progress bar text, if used, should have a mix of uppercase and lowercase text.
Presentation	Placement	Progress bar placement should complement the rest of the content.
Interaction	Animation	Progress bars should update to reflect user progression.

Prototype development

Using these preliminary design principles, the researchers developed prototypes which instantiated the design principles to allow expert focus group members to provide feedback (Winter, 2010). These prototypes of progress bars had varying color, shape, placement, and animation. The resulting prototypes consisted of four progress bar designs. The first design, Progress Bar 1 (PB 1), one was a person climbing a mountain. The second design, Progress Bar 2 (PB 2), was a circle. The third design, Progress Bar 3 (PB 3), was a shield with icons, and the last design, Progress Bar 4 (PB 4), was a lock divided into puzzle pieces (see Figure 11). All designs had cybersecurity or digital safety-related theming; with PB 1 containing a shield, PB 2

utilizing text and icons aligned with content sections, PB 3 containing a shield and social media icons to mimic badges or jewels, and PB 4 having a lock. Each progress bar was animated to demonstrate progression through the content. PB 1 showed progress through movement (person climbing a mountain) and color (progression from red to green) (see Table 5). PB 2 showed progress through progression tied to each section of content and transitioned from black and white to color through the activation (colorization) of different rings of the circle and icons. PB 3 also used the transition of color to indicate movement through the colorization of black and white elements as the user progressed. PB 4 had the puzzle pieces move from a scattered to an assembled state to indicate progress.

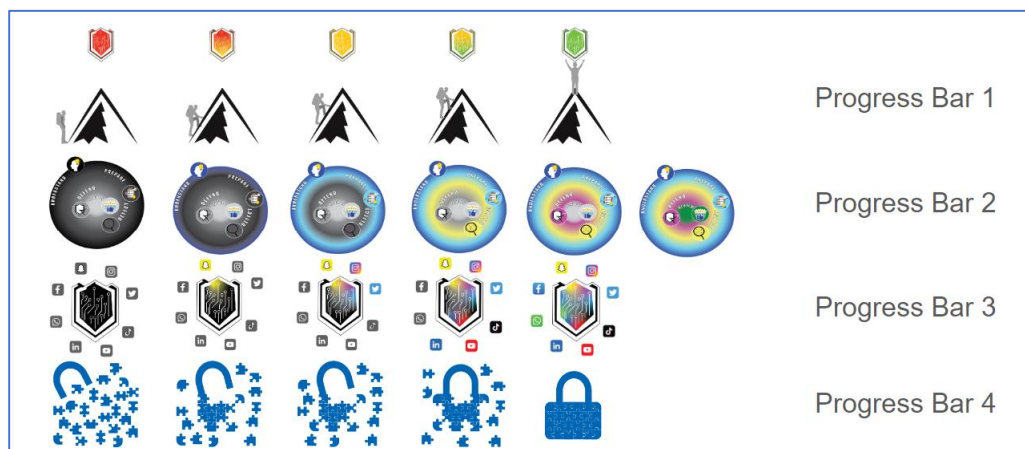


Figure 11: *Proof of Concept Designs*

These designs were evolved from the early design principles described above and served as instantiations to support different design principles. PB 1 had a tight alignment with the colors and content of the educational module. PB 2 contained text created to complement the bar. PB 3 updated the most often, and PB 4 provided the most simplistic color palette.

Table 5. *Alignment of Progress Bar Design with Initial Draft of Design Principles*

	Colors complement the rest of the content	Avoid jarring color schemes	Placement complements the rest of the content	Relate to the rest of the content	Text should have a mix of case	Update to reflect user progression
PB 1	X		X	X		
PB 2			X		X	
PB 3			X			X
PB 4		X	X			

Round 1 feedback

In order to collect feedback on the prototypes, the researchers recruited a focus group of users with backgrounds as curriculum developers, educators, and UX designers. Invitations were sent to approximately 40 individuals meeting the background requirements, with nine participants agreeing to participate in the study. As this number was in keeping with the number of participants suggested by Litosseliti (2003), the focus group proceeded, with the participants being offered two different opportunities to meet. This allowed the participants to pick a time that worked with their schedule. In the first instance, five people attended, with two being a curriculum designers and three being educators. In the second instance, four people attended, with three being UX designers and one being a curriculum developer. Focus group members used the condensed version of the UEQ, the UEQ-S, to provide feedback and to work through the digital representations of the design.

The UEQ-S was used as a topic guide, and one of the researchers served as moderator. Users were asked to introduce themselves and then talked through their responses to the UEQ-S. Open-ended questions about each UEQ-S item started as general and became more specific.

Each item on the UEQ-S was assigned a weight, ranging from 1 to 7, which was presented to participants in the form of a Likert scale. Seven was the highest level for each item. The instantiation with the highest overall UEQ-S was PB 1, which was shown to be the most supportive, easy, efficient, and clear. PB 1 had a weighted average of 5.85. PB 2 had the lowest average among the progress bars and did not receive a top score in any of the eight categories. PB 3 was seen as the most exciting, and had the second highest overall score among participants. PB 4 received top scores in inventiveness and being leading edge, but only scored third overall. Table 6 provides the results of the scores.

Table 6. Overall Round 1 Results Using UEQ-S

Characteristic	PB 1	PB 2	PB 3	PB 4	Average
Supportive	6.56	3.56	5.56	4.22	4.98
Easy	7.00	2.89	5.67	5.22	5.20
Efficient	6.67	3.33	5.78	4.67	5.11
Clear	6.67	3.38	6.56	4.33	5.24
Exciting	5.00	4.67	5.56	5.44	5.17
Interesting	5.00	4.56	5.44	5.22	5.06
Inventive	5.11	5.00	5.11	5.44	5.17
Leading Edge	4.78	4.89	5.00	5.33	5.00
Average	5.85	4.04	5.59	4.98	5.11

Curriculum developers viewed PB 1 as the most supportive, easy, efficient, and clear, and PB 4 as the most exciting, interesting, inventive, and leading edge (see Table 7).

Table 7. Curriculum Developers' Focus Group Results Using UEQ-S

Characteristic	PB 1	PB 2	PB 3	PB 4	Average
Supportive	6.56	4.00	5.00	4.33	4.97
Easy	7.00	3.67	5.00	5.00	5.17
Efficient	6.67	3.67	4.67	4.67	4.92
Clear	6.67	4.00	6.33	4.67	5.42
Exciting	5.00	4.67	5.00	5.67	5.09
Interesting	5.00	4.67	4.67	5.33	4.92
Inventive	5.11	4.00	4.67	5.67	4.86
Leading Edge	4.78	4.33	4.67	5.67	4.86
Average	5.85	4.13	5.00	5.13	5.03

Educators preferred PB 1 for all the characteristics except for being exciting and leading edge, where PB 3 was rated the most exciting and tied with PB 1 as the most leading edge (see Table 8).

Table 8. Educators' Focus Group Results Using UEQ-S

Characteristic	PB 1	PB 2	PB 3	PB 4	Average
Supportive	7.00	4.33	6.33	5.33	5.75
Easy	7.00	2.67	6.33	5.33	5.33
Efficient	7.00	3.33	6.33	5.33	5.50
Clear	7.00	2.67	6.67	5.33	5.42
Exciting	6.00	3.67	6.33	5.00	5.25
Interesting	6.00	4.33	5.33	5.33	5.25
Inventive	6.00	5.00	5.67	5.00	5.42
Leading Edge	6.00	5.00	6.00	4.67	5.42
Average	6.50	3.88	6.12	5.17	5.42

The UX designer participants rated PB 3 as the preferred design. However, they found PB 1 more supportive and easy and ranked it the same as PB 3 as the two most efficient. PB 2 and PB 4 were the most exciting. PB 2 was viewed as the most inventive. PB 3 was seen as the clearest and most interesting. In addition to being seen as exciting, the UX designers also saw PB 4 as the most leading edge (see Table 9).

Table 9. *UX Designers' Focus Group Results Using UEQ-S*

Characteristic	PB 1	PB 2	PB 3	PB 4	Average
Supportive	6.00	2.33	5.33	3.00	4.17
Easy	7.00	2.33	5.67	5.33	5.08
Efficient	6.33	3.00	6.33	4.00	4.92
Clear	6.00	3.67	6.67	3.00	4.84
Exciting	3.33	5.67	5.33	5.67	5.00
Interesting	3.67	4.67	6.33	5.00	4.92
Inventive	4.33	6.00	5.00	5.67	5.25
Leading Edge	3.33	5.33	4.33	5.67	4.67
Average	5.00	4.13	5.62	4.67	4.85

Based on focus group results, PB 1 was selected for further improvement and to further inform on the progress bar design principles.

During the discussion section of the focus group meetings, participants provided specific feedback regarding each progress bar's design as well as placement within the overall module (upper right, lower right, upper left, and lower left). Focus group feedback regarding PB 1, as well as general feedback about overall design, was collected to be incorporated in future iterations of the artifact and in the updated design principles.

Updated design principles

Round 1 feedback was classified into five categories: color, shape, text, placement and animation. This aligned with the previous progress bar characteristics identified by researchers, with the addition of overall design emerging as a theme through focus group discussion.

For colors, participants suggested integrating more green and blue to make the mountain appear to be more realistic, supporting the preliminary design principle that the color should complement the rest of the content. They also recommended that sunshine be added to the final phase of the bar to make the end look more celebratory. In order to accommodate for color blindness, one participant suggested transitioning from gray to color. These suggestions aligned with existing design principles.

Participants recommended placement for PB 1 to be in the top left or bottom right, with a stronger preference for the bottom right. Additionally, three of the participants noted that the mountain element was too big and the space between the top of the mountain and the shield was too large. These reiterated the preliminary design principle tied to placement and highlighted the

need for design principles to address the size of the bar. Therefore, researchers added: *Progress bar size should complement the rest of the content.*

For animation, participants noted that there was an unclear number of steps between the start and the finish, and for overall design, the participants suggested making both the mountain and mountain climber more realistic in appearance. This suggested a need for the design principles to address progress bar segmentation and resulted to the following being added to the design principle: *Progress bars should be segmented to reflect the content.*

In order to support the validity of the findings, member-checking was used. Participants were asked to review the findings and provide their general thoughts as well as their opinions regarding the accuracy of the findings. Participants were also invited to provide additional details to better capture their experience and feedback as to any details they thought should be removed (McKim, n.d.). The researchers made the following refinements to the design principles, noted in bold, based on Round 1 feedback (see Table 10):

Table 10. Design Principle Draft Based on Round 1 Feedback

Element	Characteristic	Preliminary Design Principle	Updated Design Principle (changes in bold)
Representation	Color	Progress bar color should complement the rest of the content but should be distinct and recognizable.	Progress bar color should complement the rest of the content but should be distinct and recognizable.
	Color	Progress bar designers should avoid jarring color schemes and ensure colors meet the needs of users with color blindness or low vision.	Progress bar designers should avoid jarring color schemes and ensure colors meet the needs of users with color blindness or low vision.
	Shape	Progress bar design should relate to the rest of the content.	Progress bar design should relate to the rest of the content.
	Text	Progress bar text, if used, should have a mix of uppercase and lowercase text.	Progress bar text, if used, should have a mix of uppercase and lowercase text.
Presentation	Placement	Progress bar placement should complement the rest of the content.	Progress bar placement should complement the rest of the content.
	Placement		Progress bar size should complement the rest of the content.
Interaction	Animation	Progress bars should update to reflect user progression.	Progress bars should update to reflect user progression.
	Animation		Progress bars should be segmented to reflect the content.

To incorporate these new design principles, PB 1 was improved and integrated into a cybersecurity learning module, *Steganography: A Cybersecurity Adventure*. Based on Round 1 feedback, both PB 1 and a traditional progress bar were placed in the bottom right corner of the module for comparison. The size of PB 1's elements was adjusted, along with the coloration. PB 1 was adjusted so that the mountain scene appeared more realistic, and colors adjusted from grayscale to color as the participant progressed through the module. The final stage of PB 1 included sunshine being added to make it appear more celebratory. In addition to having the design progress from gray scale to color to better define the stages, the researchers also incorporated stepping-stones to further define the stages and to accommodate for better segmentation.

Round 2 feedback

After refining PB 1 to represent updated design principles, the researchers placed the instantiation into a functioning online learning environment. The researchers provided two progress bar versions, PB 1 and a traditional progress bar, to college students for comparison for Round 2 feedback. During this second round of feedback, users were presented with both progress bars integrated into identical online learning environments.

College students were asked to compare two different versions of a cybersecurity module. Version A contained a traditional progress bar (one with a rectangle shape). Version B contained the refined version of PB 1, which was preferred by the focus group during Round 1. These two versions were used to determine commonalities in design feedback to inform on design principles.

Both versions of the learning module were presented to all participants. The first, Version A, contained the traditional progress bar. The second, Version B, contained PB 1 (see Figure 12). The versions were distributed among graduate and undergraduate students at a university located in the southeastern United States via a Google Form.

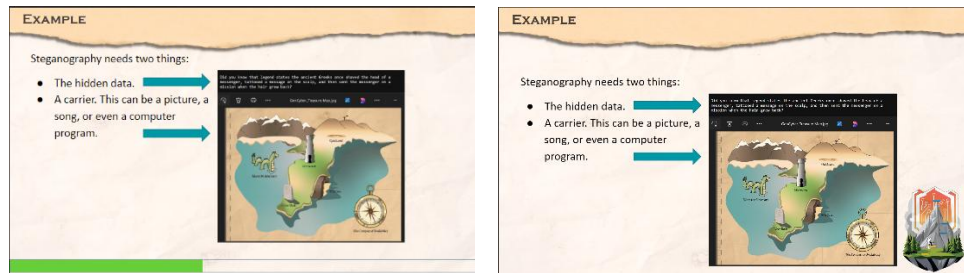


Figure 12: Screenshots of Version A (left) and Version B (right)

After reviewing Version A and Version B, participants were asked to determine each version's alignment with the desired characteristic associated with the progress bar. A Likert scale was given to participants to provide this alignment. The scale ranged from 1 to 7, with 7 indicating the most alignment with the specified characteristic. The characteristics included supportive of UX, easy to understand, efficient, clear, exciting, interesting, inventive, and leading edge. Participants were then invited to justify their answers for each selection through open-ended questions.

The UX data collected using the surveys provided insight as to the perceived strengths and weaknesses for the traditional progress bar and the hedonic progress bar. The researchers used this feedback to identify categories regarding user benefits and concerns and to infer themes related to the design of the bars. The themes were then used for design principle refinement (Saldaña, 2013).

Before evaluating the responses, a preliminary analysis of the data was completed to determine if there were any missing values, outliers, or suspicious response patterns. Of the 42 responses, one student had left the survey before providing any responses. This participant's data, a series of blank responses, was removed from the data set, leaving 41 responses for analysis.

When examining the close-ended questions' results of the comparison between Version A, the traditional progress bar, and Version B, the hedonic progress bar; Version B had a higher score for all the characteristics combined, with a mean score of 4.48 (4.5 median) out of a possible 7 points. Version A had an overall mean score of 4.37 (4.38 median) out of a possible 7 points (see Figure 13). The standard deviation for Version B was 1.09 and for Version A was .84.

Characteristics Combined

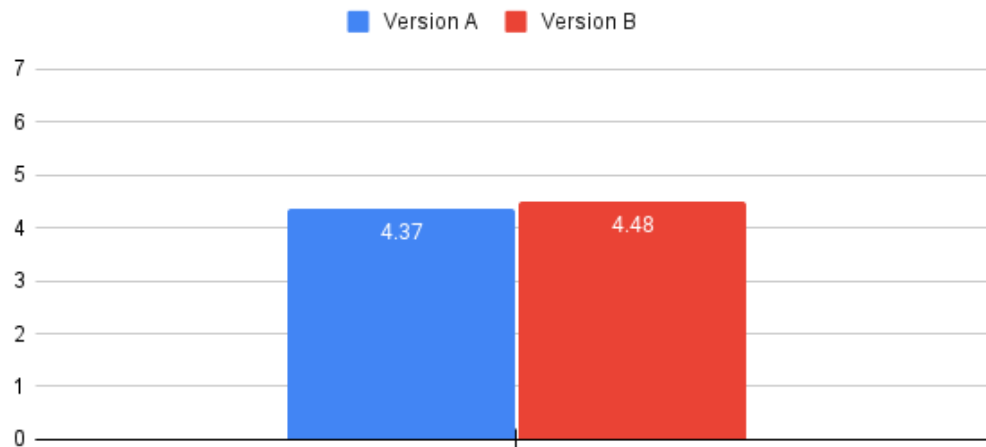


Figure 13: Comparison of Overall Characteristics Ratings

A breakdown of these overall characteristic scores by user can be seen in Figure 14. When averaging the characteristics of Version A per user and comparing it to Version B, Version B had a higher average for 21 users, with 51% of users giving it a higher score, while Version A had a higher average for 19 users, with 46% of users giving it a higher score. There was one instance where Version A and Version B had the same average for a single user.

Comparison of Combined Characteristics by User

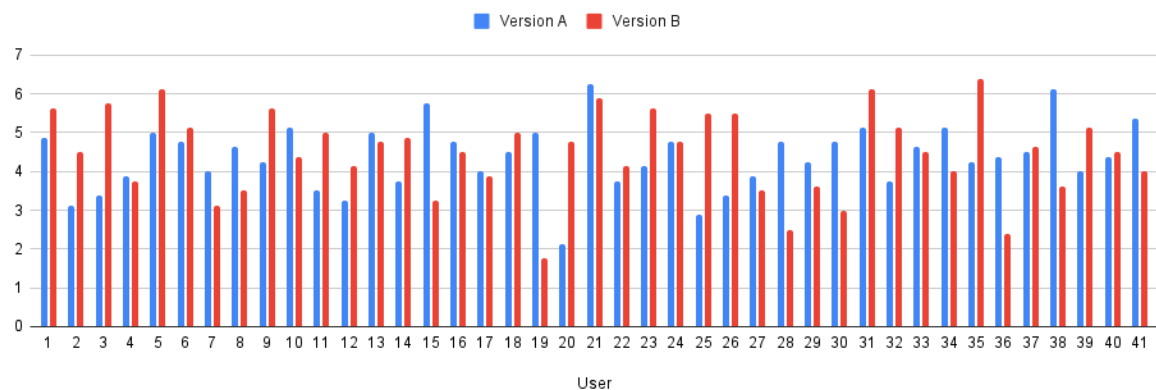


Figure 14: Comparison Overall Characteristics Ratings by User

When looking at individual characteristics, the users found Version A to be on average more supportive, easy to understand, efficient, and clear; however, the users found Version B to be more exciting, interesting, inventive, and leading edge (see Figure 15).

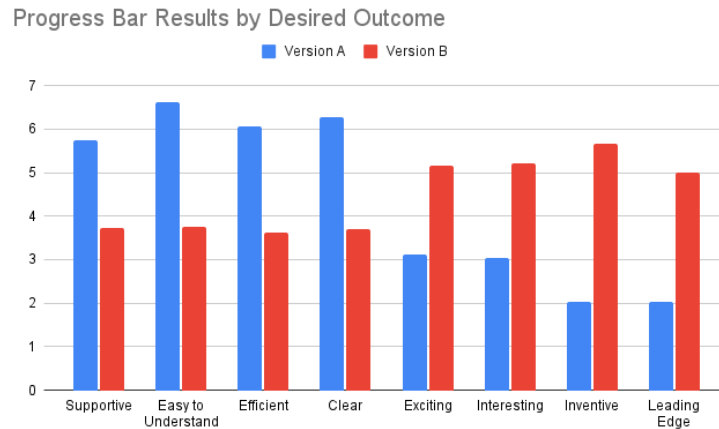


Figure 15: Average User Ratings per Characteristic

In the area of support, Version A had a mean of 5.75 and a median of 6. Version B had a mean of 3.73 and a median of 4. As far as finding the progress bars easy to understand, Version A had a mean of 6.61, with a median of 7, while Version B had a mean of 3.76, with a median of 3. The ratings were similar for efficient and clear. For efficiency, Version A had a mean of 6.073, with median of 6; and Version B had a mean of 3.61 and a median of 3. Version A had a mean of 6.23 and a median of 7 for being clear, while Version B had a mean of 3.71 and a median of 4 for being clear.

As noted earlier, users found Version B to be more exciting, interesting, inventive, and leading edge. Here, Version A had a mean of 3.122 and a median of 3 for exciting, while Version B had a mean of 5.17 and a median of 5. Version A's mean for interesting was 3.05, with a median of 3, and Version B's mean for interesting was 5.22, with a median of 6. When considering inventiveness, users saw Version A as having little to no inventiveness, with it having a mean of 2.02 and a median of 1. However, Version B was rated as very inventive, with a mean of 5.67 and a median of 6. The results were similar when users considered the characteristic of being leading edge. In this case Version A had a mean of 2.02 and a median of 1, and Version B had a mean of 5 and a median of 5. These results; as well as information regarding standard deviation, excess kurtosis, and Cramer-von Mises p value; are provided in Table 11.

Table 11. Round 2 Survey Results for Progress Bar Versions A and B

Characteristic	Mean	Median	Standard deviation	Excess kurtosis	Skewness	Cramér-von Mises p value
Clear_Version A	6.268	7	1.326	5.887	-2.346	0
Clear_Version B	3.707	4	1.742	-1.012	0.126	0.002
Easy to Understand_Version A	6.61	7	0.793	4.311	-2.208	0
Easy to Understand_Version B	3.756	3	1.91	-1.172	0.255	0.008
Efficient_Version A	6.073	6	1.156	2.468	-1.526	0
Efficient_Version B	3.61	3	1.666	-0.849	0.261	0.013
Exciting_Version A	3.122	3	1.383	-0.787	0.232	0.006
Exciting_Version B	5.171	5	1.496	0.121	-0.712	0.006
Interesting_Version A	3.049	3	1.306	-0.866	-0.025	0
Interesting_Version B	5.22	6	1.554	0.974	-1.111	0
Inventive_Version A	2.024	1	1.538	1.458	1.587	0
Inventive_Version B	5.659	6	1.027	0.415	-0.657	0
Leading Edge_Version A	2.024	1	1.538	2.164	1.67	0
Leading Edge_Version B	5	5	1.189	1.873	-0.903	0
Supportive_Version A	5.756	6	1.494	1.269	-1.339	0
Supportive_Version B	3.732	4	1.781	-0.796	0.235	0.063

All the p values for the characteristics except one, Supportive_Version B, are well below .05, showing a statistically significant difference among the means of the progress bars' characteristics. These results are in line with the alignment between motivational mechanics and UX discussed in chapter 3, with the traditional design being shown to appeal to participants' cognitive perspective and rating higher in the pragmatic aspects of UEQ (perspicuity, efficiency, and dependability) and the hedonic design being shown to better support the perspective of interest and rating higher in the hedonic aspects of UEQ (stimulation and novelty) (Sailer et al., 2013; Schrepp et al., 2017).

While the closed-ended results informed on each bar's ability impact UX and alignment with motivational mechanics, the open-ended results allowed the researchers to identify categories regarding user benefits and concerns. This helped the researchers to refine design principles to impact motivation. In the open-ended sections, participants were asked to explain their choice for each close-ended survey item.

These responses were uploaded into ATLAS.ti Web for coding and analysis. This tool was beneficial in storing, organizing, managing, and reconfiguring data (Saldaña, 2013). The data was first coded by UX category and the progress bar version (A for traditional or B for hedonic) discussed. This preliminary coding allowed the researchers to apply filters to better understand coding trends. After doing this initial coding, the researchers next examined the data multiple times, looking for core content that could be deemed significant to progress bar design (Patton, 2002). This included sorting the data by progress bar version and by UX categories to

look for trends related to the progress bar types or by the users' experience. For example, the researchers noticed quotes like "I liked the connection between finishing the tasks and scaling the mountain," which a participant noted in relation to inventiveness, and "The mountain was more visually appealing," which a different participant connected with interesting. These, and other quotes related to the shape of the hedonic progress bar, began to emerge and connect together as concepts and later as categories. Axial coding was used to sort and relabel synonyms and to remove redundant codes (Saldaña, 2013). For example, codes like "too complex" or "complicated" were later consolidated under another code, "design complexity."

The following concepts emerged: progress bar size, progress bar design complexity, progress bar color, progress bar shape, and progress bar segmentation. Codes related to these concepts were reviewed again; this time, the researchers captured the following information:

- number of different users to help understand the commonality of the comments;
- associated close-ended survey item to understand the item that prompted the comment;
- the version of the progress bar associated with the comment; and
- if the comment was a positive or negative comment.

There were 12 references to progress bar size from nine different users. Of those 12 references, 3 were associated with progress bar support, 2 were associated with the clearness of the progress bars, 2 were associated with the progress bars being leading edge, 1 reference to the progress bars' efficiency, and 1 reference to the progress bars' inventiveness. In 3 different instances users commented on the size of the progress bars in responses that were not tied to a characteristic. When discussing the size of the bar, the traditional shape's size was mentioned 3 times, and the hedonic bar's size was mentioned 9 times. Two of the comments about the traditional shape's size were positive, and one was neutral. All the references to the hedonic bar's size were negative. This led to two categories: 1. Progress bar size should complement the design of the progress bar, and 2. Progress bar size should complement the rest of the content.

Participants made 17 references to the design's placement. These were related to clear (5), support (3), easy to understand (2), interesting (2), exciting (2), efficient (1), inventive (1), and leading edge (1). These references came from 15 distinct users. Most of the comments linked to placement of the progress bars were tied to the hedonic progress bar, with only 3 of the comments referencing the traditional progress bar. As with the concept of progress bar size, all 9

of the negative comments were tied to the hedonic progress bar; however, there were 3 positive comments tied to the hedonic progress bar. This led to two categories: 1. Progress bar complexity should scale well across varying screen sizes and display formats, and 2. Progress bar design should match the content's theme.

In regard to color, there were 18 references, from 15 different users. Participants referred to color when asked about the progress bars being clear (7), interesting (6), exciting (2), supportive (1), and efficient (1). One reference was to a question not associated with a characteristic. Half of the comments were tied to the traditional progress bar and half were tied to the hedonic progress bar. Both negative comments were tied to the traditional progress bar, which stemmed from the color green being too bright and the use of only one color being unimaginative. Participant feedback led to two categories: 1. Progress bar colors should be bright enough to be easy to see and distinct from other content but not so bright that they are distracting and 2. Progress bars utilizing multiple colors can be eye catching and motivating.

A total of 15 participants commented on the shape of the progress bars through a total of 20 references. All but two of the references were positive, with the negative comments involving the rectangle shape of the traditional progress bar being boring. The shape of the hedonic progress bar was noted 16 times by participants, and all those comments were positive. Participants mentioned the shape of the bar when explaining their choices regarding the bars' being interesting (7), exciting (5), leading edge (3), clear (1), easier to understand (1), inventive (1), and supportive (1). There was 1 reference to shape that came from a question not directly tied to a characteristic. Three categories were the result of this feedback: 1. Progress bar shape should connect with the content, 2. Progress bar shape should be visually appealing, and 3. Progress bar shape should be easy to understand.

Animation was the most referenced topic related to progress bar design, receiving 30 comments from 22 of the participants. Segmentation was mentioned in relation to the progress bars being supportive (6), easy to understand (5), clear (4), inventive (3), efficient (1), exciting (1), and interesting (1). Participants provided 9 comments about the segmentation in a response that was not tied to a characteristic. A total of 20 of the comments involved the hedonic progress bar, and 11 of the comments were negative, with all negative comments involving the hedonic progress bar. The negative comments stemmed from feelings that the mountain less accurately portrayed the actual time left in the module, as the milestones were placed equally up the

mountain path and the spacing did not capture the time needed to complete the subsection as opposed to other subsections. Four of the commenters also noticed that the animations associated with the completion of each segment were too slow. Three participants commented that mountain segmentation (a hiker climbing the mountain) was motivating, and two more participants noted that using a hiker progressing through milestones to indicate progress made the module gamified and closely aligned with the content. Three categories were the result of this feedback: 1. Progress bar segmentation, when aligned to content, motivates users, 2. Progress bars should update frame by frame to better reflect user progression, and 3. Progress bar segmentation should be representative of the content remaining in the task. These results are summarized in Table 12. Table 13 provides a summary of the concepts and categories.

Table 12. *Concepts tied to UX Characteristics*

Characteristic	Size	Design Complexity	Color	Shape	Segmentation
Clear	2	5	7	1	4
Easier to Understand	0	2	0	1	5
Efficient	1	1	1	0	1
Exciting	0	2	2	5	1
Interesting	0	2	6	7	1
Inventive	1	1	0	1	3
Leading Edge	2	1	0	3	0
Supportive	3	3	1	1	6
General/Other	3	0	1	1	9
Total	12	17	18	20	30
References					

Table 13. *Categories Aligned with Concepts*

Concepts	Categories
Size	1. Progress bar size should complement the design of the progress bar and the rest of the content. 2. Progress bar size should complement the rest of the content.
Placement	1. Progress bar complexity should scale well across varying screen sizes and display formats 2. Progress bar design should match the content's theme.
Color	1. Progress bar colors should be bright enough to be easy to see and distinct from other content but not so bright that they are distracting. 2. Progress bars utilizing multiple colors can be eye catching and motivating.
Shape	1. Progress bar shape should connect with the content. 2. Progress bar shape should be visually appealing. 3. Progress bar shape should be easy to understand.
Animation	1. Progress bar segmentation, when aligned to content, motivates users. 2. Progress bars should update frame by frame to better reflect user progression. 3. Progress bar segmentation should be representative of the content remaining in the task.

The researchers aligned these categories with the progress bar elements of representation, presentation, and interaction, which are discussed in chapter 3.

Final design principles

Following the analysis of user feedback from Round 2, the researchers classified the emerging categories by the identified progress bar elements of representation, presentation, and interaction. As discussed in chapters 3 and 5, *representation* is the selection of techniques to illustrate important concepts and is tied to the characteristics of object color, shape, and text. *Presentation* is the space, time, and human visual system and is tied to considerations related to object placement. *Interaction* is the coupling of input and output and is associated with object animation (Spence, 2014).

Classification of Round 2 feedback categories by design elements

The classification of the categories pulled from user feedback using these progress bar elements was the first step in aligning these categories with the earlier drafts of the design principles.

Representation

The following categories were aligned to representation:

- *Progress bars utilizing multiple colors can be eye catching and motivating.*
- *Progress bar colors should be bright enough to be easy to see and distinct from other content but not so bright that they are distracting.*
- *Progress bar size should complement the design of the progress bar and the rest of the content.*
- *Progress bars should be visually appealing.*
- *Progress bars should be easy to understand.*

These categories are further described below.

Progress bars utilizing multiple colors can be eye catching and motivating.

Users appreciated the use of multiple colors. One user noted, “The transition of color as you progress is very clear.” Another noted that it was “nice seeing the colors change as the user reaches milestones.” One user stated, “The colors were nice to look at and it painted a great

motivational picture that strangely gave me a bigger feeling of accomplishment.” These comments emphasize the importance of color use.

Progress bar colors should be bright enough to be easy to see and distinct from other content but not so bright that they are distracting.

Another reoccurring theme also dealt with the use of color. Designers and developers should be sure that the colors are eye catching without being distracting. Users appreciated that the colors used stood out and were eye catching. However, one user noted that the green used for the traditional progress bar was too bright, and the user reported having trouble staying focused. Another felt the color was “a little bit off putting.” These comments reinforce the need for developers and designers to be intentional with color choices to make sure the colors are distinct, yet complementary to other design elements.

Progress bar size should complement the design of the progress bar and the rest of the content.

Expectations regarding progress bar size were reflected through comments such as “It serves its purpose without taking up much space or being distracting.” Other commenters critiqued the size of one of the artifacts. They noted issues seeing progress between the milestones and having to zoom in to check progress. These comments emphasize the need for designers and developers to ensure that the size of the progress bar is sufficient to be easily understood by the user; however, the size should not be so large it is in the way of other content.

Progress bars should be visually appealing.

Users made several comments regarding the shape of the progress bar, highlighting the need for the shape to be visually appealing. Users’ comments on this topic centered around the hedonic progress bar. For example, one participant noted, “Having never seen something like bar B [the hedonic progress bar] . . . it offers a bigger sense of accomplishment.” Another participant stated, “The mountain was more visually appealing.” Multiple users indicated that they found the shape of the hedonic progress bar interesting, with one user noting that “It looks like more care was put into it.” These comments contrast with comments about the shape of the traditional progress bar. Users saw the traditional progress bar as boring and flat. These comments bring to light the importance of the progress bar being visually appealing.

Progress bars should be easy to understand.

Users noted an appreciation for being able to quickly and easily interpret the percent of tasks remaining in the module. They mentioned a need for “a quick and clean representation” of progress in the module. One user noted that the hedonic design was difficult to read, while another user commented that it took an extra second to process and interpret. Users valued the ability to read the information “with a quick glance or even in peripheral vision.” Therefore, designers and developers should take into consideration the users’ ability to understand the information being displayed.

Presentation

Researchers identified the following categories related to presentation:

- *Progress bar design should complement and connect with the rest of the content.*
- *Progress bar complexity should scale well across varying screen sizes and display formats..*

A couple of themes that were prevalent across much of the participant feedback revolved around the progress bar’s alignment with content. This alignment was associated with the context and detail of the progress bars and, therefore, tied to the element of presentation. Some comments dealt with the overall look and feel of the content. Other comments dealt with how well the progress bar was integrated in relationship with other elements on the screen. The themes are summarized below.

Progress bar design should complement and connect with the rest of the content.

As discussed earlier, the hedonic progress bar was designed to connect with a jungle adventure theme, with the progress bar being a mountain. Several participants noted enjoying the connection. One participant stated, “Bar A [the traditional progress bar] I’ve seen before while this was the first time I saw something like Bar B. I liked the connection between finishing the tasks and scaling a mountain.” Another participant noted, “It [the hedonic progress bar] offers a bigger sense of accomplishment when milestones are completed by having the little person make their way up the mountain. Bar A [the traditional progress bar] really just tells me how much I have left.”

Other participants tied the connection with the content directly to motivation. For example, one participant stated, “It [the hedonic progress bar] made me want to complete the task and finish climbing the mountain.” Another participant echoed this idea by stating, “Watching the climber go up the mountain adds an element of excitement and a feeling of

accomplishment to the lesson. It reinforces the adventure theme.” Another participant stressed the benefits of a customized design; the participant noted, “It felt individualized to the training and was a great motivator. It felt like it was made especially for the training. That being said, if the company I was working for gave me training to do and it had a progress bar like that, I would be more inclined to believe that the company cared about the training and what I got out of it.”

Progress bar complexity should scale well across varying screen sizes and display formats.

In addition to allowing for opportunities to tie the design to the rest of the content thematically, designers and developers should also consider how the progress bar functions in relation to the rest of the content being displayed. Participants noted that the mountain shape of the hedonic progress bar did not scale well. They mentioned that it was too small on the screen to see all the details. Participants used phrases such as “difficult to read” and “too cluttered.” This was especially noted with participants who were using their phones. This highlights the importance of making sure designs translate well across various display types and sizes.

Interaction

Finally, the researchers identified the categories that related to interaction. These were

- *Progress bars should update frame by frame to better reflect user progression.*
- *Progress bar segmentation should be representative of the content remaining in the task.*

In this case, interaction refers to the animations and/or filling of the bar that indicates user progress. Two design principles surfaced regarding artifact interaction.

Progress bar behavior reflects the advancement of user progress and is instrumental in motivating the user. The following is a detailed explanation of the two principles tied to progress bar behavior.

Progress bars should be updated frame by frame to better reflect user progression.

Participants stressed the need for the progress bar to continuously update in order to display up-to-date user progression. Users expressed a desire to see the progress bar update frame by frame, allowing them to see the impact of their work within the module. Participants noted feeling stuck and not knowing where they were when the progress bar did not update continuously.

Progress bar segmentation should be representative of the content remaining in the task.

Participants also expressed the need for the progress bar's segmentation to accurately reflect the percentage of work completed and left to be completed within the module. One user noted, "It [spacing the movement of the progress bar] felt arbitrary about the size of each step." Other users expressed concerns about it being difficult to understand progress and it not being obvious how much was left.

These comments capture the need for progress bars to update continuously and to accurately represent the percent of content remaining.

Table 14 provides a summary of the alignment of progress bar elements to characteristics identified from user feedback.

Table 14. *Classification of Round 2 Feedback Categories by Design Elements*

Design Element	Category from Round 2 Feedback
Representation	Progress bars utilizing multiple colors can be eye catching and motivating. Progress bar colors should be bright enough to be easy to see and distinct from other content but not so bright that they are distracting. Progress bar size should complement the design of the progress bar and the rest of the content. Progress bars should be visually appealing. Progress bars should be easy to understand.
Presentation	Progress bar design should complement and connect with the rest of the content. Progress bar complexity should scale well across varying screen sizes and display formats.
Interaction	Progress bars should update frame by frame to better reflect user progression. Progress bar segmentation should be representative of the content remaining in the task.

Gap analysis

The researchers next compared the design principles identified through literature and focus group feedback (see Table 10) with the categories identified through user feedback (see Table 13) to look for gaps and overlaps. This approach is similar to one used by Marx et al. (2011), who reconciled survey data with existing knowledge to identify gaps in design principles. The alignment between the design principles and focus group feedback and with the ones that emerged from user feedback is shown in Table 15.

Table 15. Comparison of Design Principles Based and Categories from Round 2 Feedback.

Element	Characteristic	Prior Draft of Design Principles	Category from Round 2 Feedback (gaps in bold)
Representation	Color	Progress bar color should complement the rest of the content but should be distinct and recognizable.	Progress bars utilizing multiple colors can be eye catching and motivating. Progress bar colors should be bright enough to be easy to see and distinct from other content but not so bright that they are distracting. Progress bar design should complement and connect with the rest of the content.
	Color	Progress bar designers should avoid jarring color schemes and ensure colors meet the needs of users with color blindness or low vision.	
	Shape	Progress bar design should relate to the rest of the content.	
	Text	Progress bar text, if used, should have a mix of uppercase and lowercase text.	
	Color, Shape, and Text Color, Shape, and Text		
Presentation	Placement	Progress bar placement should complement the rest of the content.	Progress bars should be visually appealing. Progress bars should be easy to understand.
	Placement	Progress bar size should complement the rest of the content.	
Interaction	Animation	Progress bars should update to reflect user progression.	Progress bar should update frame by frame to better reflect user progression. Progress bar segmentation should be representative of the content remaining in the task.
	Animation	Progress bars should be segmented to reflect the content.	

Researchers reviewed the gaps identified in bold in Table 15 and decided that the categories identified from user feedback should be retained. The researchers also noted two themes that did not emerge from user feedback. These were concerns about meeting the needs of

users with color blindness or low vision and the use of mixed-case text. This could be because the concerns were adequately addressed with prototype refinement or because no one in the user population experienced color blindness or low-vision. To address this difference, the researchers chose to keep principles related to these concerns. The final, updated list of design principles is shown in Table 16, with these changes reflected in bold.

Table 16. *Design Principles for Visual Progress Bars After Gap Analysis*

Element	Characteristic	Final Design Principle (changes in bold)
Representation	Color	Progress bars utilizing multiple colors can be eye catching and motivating. Progress bars should include color combinations that meet the needs of users with color blindness or low vision. Progress bar colors should be bright enough to be easy to see and distinct from other content but not so bright that they are distracting.
	Shape	Progress bar design should complement and connect with the rest of the content.
	Text	Progress bar text, if used, should have a mix of uppercase and lowercase text.
	Color, Shape, and Text	Progress bars should be easy to understand.
	Color Shape, and Text	Progress bar size should complement the design of the progress bar and the rest of the content.
Presentation	Placement	Progress bar complexity should scale well across varying screen sizes and display formats.
Interaction	Animation	Progress bars should update frame by frame to better reflect user progression.
	Animation	Progress bar segmentation should be representative of the content remaining in the task.

CHAPTER 6

RESULTS

Following the gap analysis described in the previous chapter, the researchers used the design principles to update a prototype to instantiate the design principles, adjusting the prototype used in Round 2 of the demonstration phase to account for the resulting refinements of the design principles. This chapter presents the results of the summative evaluation of the proposed design principles.

Focus group interviews were conducted with students over the course of three sessions, with two of the sessions being in-person and one being a virtual session via Zoom. Sessions were analyzed and repeated until theoretical saturation was reached, meaning that no new or relevant data seemed to emerge (Strauss and Corbin, 1998). A total of 11 students participated, with five of the students opting for an in-person session and six students attending the virtual session. Participants were all students at the same university as from the Round 2 feedback; however, students who participated in the Round 2 feedback were excluded from participating in these sessions. The students varied in academic level and academic major, with one participant being a freshman, two being sophomores, four being juniors, three being seniors, and one being a graduate student. Areas of study included cybersecurity engineering (3), information systems (2), education (1), psychology (1), history (1), computer science (1), electrical engineering (1), and nursing (1).

Focus group sessions began with a brief introduction, which also included an overview of participant eligibility and expectations. Next, the researcher discussed the purpose of the discussion and reviewed and collected the consent forms. At this point, the researcher began recording the sessions and asked participants to rate the two progress bars (traditional and hedonic) based on the indicators of UX using the shortened form of the User Experience Questionnaire (UEQ-S). Next, participants were invited to discuss their findings, with the researchers occasionally asking participants to further explain their response or inviting other participants for additional input.

Participants attending the in-person sessions were given one version of the UEQ-S survey, where the traditional progress bar (rectangle shape) was listed first and presented on the

left side of the side-by-side screen for comparison, while participants in the virtual session were given a version of the UEQ-S survey where the hedonic progress bar (mountain shape) was listed first and presented on the left side of the side-by-side screen. Other than this variation, all materials were the same.

The UX survey utilized a Likert scale ranging from 1 to 7, with 7 indicating the most alignment with the characteristics of support, ease, efficiency, clearness, excitement, interest, inventiveness, and leading edge. Participants were also provided with open-ended questions to allow them to explain their choices. For the purpose of data analysis, the traditional progress bar will be referred to as Version A, and the hedonic progress bar will be referred to as Version B as shown in Figure 16.

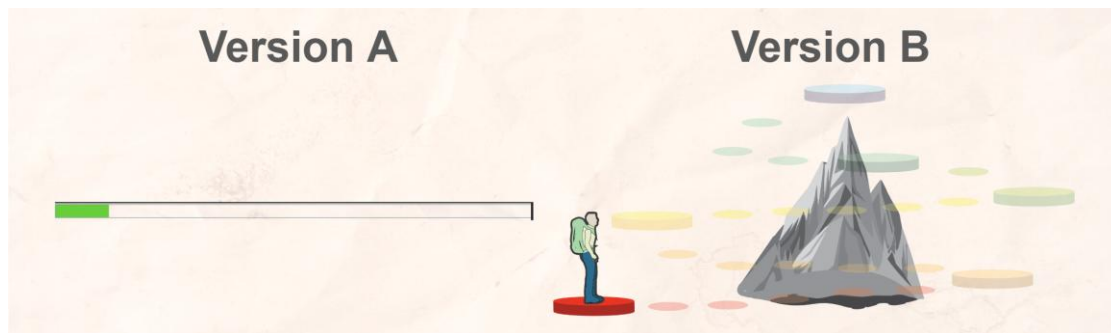


Figure 16: *Version A and Version B Used for Final Evaluation*

When looking at the characteristics combined, Version B, the hedonic progress bar, rated higher than Version A, the traditional progress bar. Characteristics aligned with Version B had a mean score of 5.95 out of a possible 7 points. Characteristics aligned with Version A had a mean score of 3.89 out of a possible 7 points (see Figure 17).

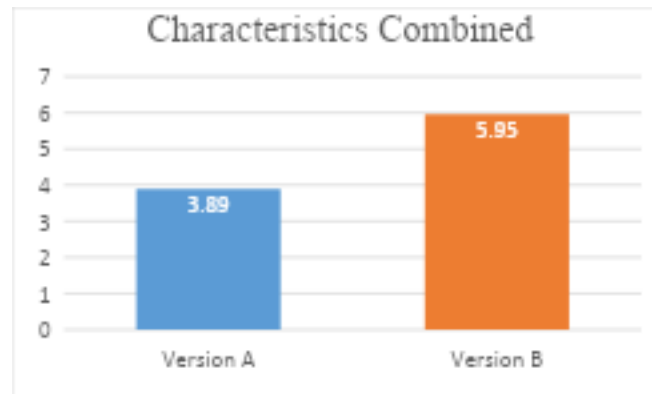


Figure 17: Average Results of the UEQ-S Characteristics for Version A and Version B

A breakdown of results from individual users can be seen in Figure 18. When looking at the average rating of the combined characteristics of Version A and Version B by each user, Version B consistently scored higher than Version A. There was no instance where Version A and Version B had the same average for a single participant.

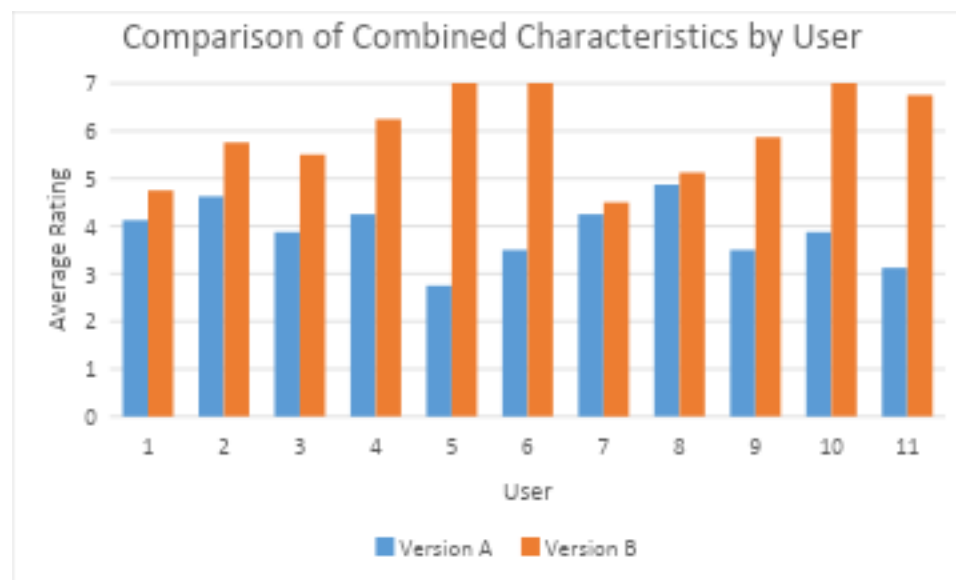


Figure 18: Average Ratings of Version A and Version B by Individual Participant

As far as the individual characteristics, Version A performed slightly better than Version B in the areas of easy to understand, efficient, and clear; however, Version B performed better on average with participants in the areas of exciting, interesting, inventive, and leading edge. Participants found Version A and Version B equally supportive (see Figure 19).

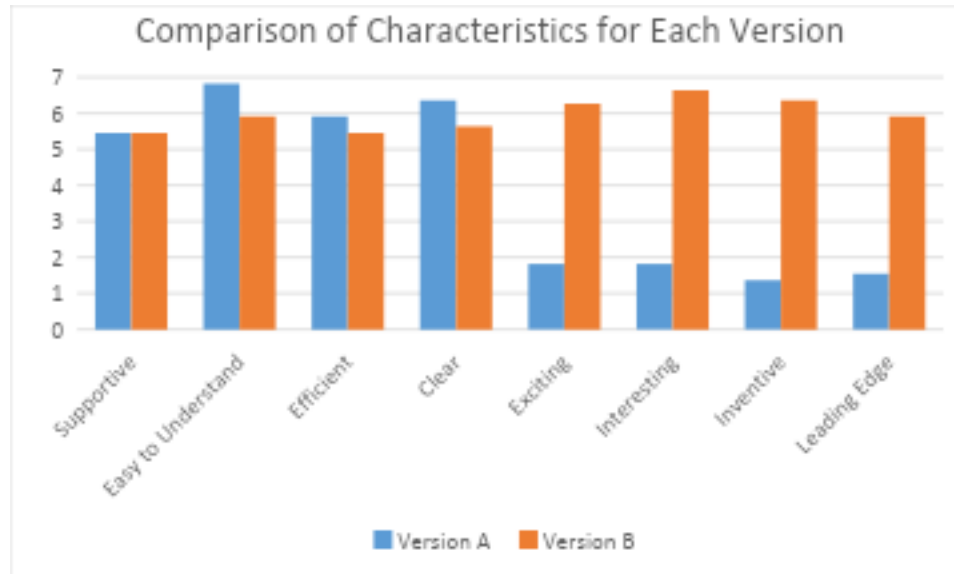


Figure 19: Comparison of Average Ratings for Version A and Version B by Characteristic

Participants' scores for versions A and B in the area of support had a mean of 5.46 out of 7, with Version A having a median of 5 and Version B having a median of 6. For being easy to understand, Version A had a mean of 6.82 (median of 7), while Version B had a mean of 5.91 (median of 6). In the area of efficiency, Version A had a mean of 5.91 (median of 6), and Version B had a mean of 5.46 (median of 6). The ratings were also very close for the characteristic of clear, with Version A having a mean of 6.36 (median of 7) and Version B having a mean of 5.64 (median of 7).

While the ratings were similar for the previous characteristics, Version B scored far higher in the next four categories. Version B had a mean of 6.27 (median of 7) for exciting, compared to Version A's mean of 1.82 (median of 1). The gap in ratings was even greater for the characteristic of interest, with Version B having a mean of 6.64 (median of 7) to Version A's mean of 1.82 (median of 1). Participants also found Version B more inventive than Version A, with means of 6.36 (median of 7) and 1.34 (median of 2), respectively. Finally, Version B was also seen as more leading edge, with a mean of 5.91 (median of 6), compared to Version A having a mean of 1.55 (median of 1). The p values for all the characteristics but Supportive_Version_A are below .05, demonstrating there are statically significant differences in the means of the progress bars' characteristics. Table 17 summarizes the results.

Table 17. *UEQ-S Survey Results from Evaluation*

Characteristic	Mean	Median	Standard deviation	Excess kurtosis	Skewness	Cramér-von Mises p value
Supportive_Version A	5.46	5	1.233	-0.351	-0.387	0.211
Supportive_Version B	5.46	6	1.725	-1.735	-0.592	0.004
Easy to Understand_Version A	6.82	7	0.386	2.037	-1.923	0
Easy to Understand_Version B	5.91	6	1.083	-1.626	-0.289	0.008
Efficient_Version A	5.91	6	1.443	4.285	-1.93	0.011
Efficient_Version B	5.46	6	1.725	-0.54	-0.839	0.042
Clear_Version A	6.36	7	1.15	7.03	-2.536	0
Clear_Version B	5.64	7	1.772	-0.307	-1.062	0.003
Exciting_Version A	1.82	1	1.192	3.867	1.912	0.002
Exciting_Version B	6.27	7	1.135	0.877	-1.507	0
Interesting_Version A	1.82	1	1.192	3.867	1.912	0.002
Interesting_Version B	6.64	7	0.643	2.611	-1.8	0
Inventive_Version A	1.36	1	0.881	8.042	2.808	0
Inventive_Version B	6.36	7	1.068	0.427	-1.41	0
Leading Edge_Version A	1.55	1	0.988	2.449	1.833	0
Leading Edge_Version B	5.91	6	1.083	-1.626	-0.289	0.008

Open-ended responses and transcripts from participant interviews were coded and analyzed using ATLAS.ti. The final, combined data set consisted of 11 open-ended responses as well as transcripts from three focus group interviews. For each focus group session, open coding was first used to ensure a close examination of data and to compare and group concepts (Smit, 2002). The researchers coded the data until saturation was reached (O'Reilly et al., 2012). Next, the researchers combined the data sets in order to utilize selective coding (Strauss and Corbin, 1998). Selective coding is a way to examine interconnectedness among categories (Rabinovich and Kacen, 2010). Across the combined data set, there were a total of 54 instances of coding, aligned to 7 code groups. Of those instances, 17 were tied to motivation, 10 were tied to complementing and connecting with content, 8 were linked to animation, 7 referenced color, 6 were in relation to shape, 4 concerned ease of understanding, and 2 were in relation to size. Table 18 summarizes these code groups.

Table 18. *Overview of Codes per Code Group*

Code Group	Motivation	Complementing and Connecting with Content	Animation	Color	Shape	Ease of Understanding	Size
Instances of Coding	17	10	8	7	6	4	2

The coding instances were further aligned to the proposed design principles as well as overall participant motivation in order to understand the usefulness of the design principles to inform progress bar designers and developers on how to design progress bars to improve motivation. As far as overall comments regarding motivation, the majority of the comments, all but 4, were tied to the hedonic progress bar. The comments in reference to the hedonic progress bar included descriptors such as new, exciting, fun, accomplishment/success, and immersive. Example comments included: “It’s new and exciting,” “The way the Progress Bar B is set up feels almost like the level maps in video games, which made it much more exciting than the traditional progress bar,” “I think the mountain climber is most fun,” “It gave me a sense of accomplishment to receive shields,” and “The mountain feels like a more immersive experience and that is really a way to keep future students engaged in the material.” The other comments, which centered around Version A, dealt with liking the simplicity and familiarity of the traditional progress bar. When examining comments that tie back to progress bar design, coding instances were aligned with design principles to determine if the design principle was supported.

Version B, which was an instantiation of the design principles, used color to indicate the various segments of the learning module, with the first section represented by red stepping stones, the second section represented by orange stepping stones, the third section represented by yellow stepping stones, and the fourth section represented by green stepping stones. At the beginning of the module, all stepping stones were transparent in cover, but they filled in as the user progressed through the module. The hiker, a characterized element of the progress bar, was also colorized, being portrayed as having blue pants, a cream shirt, and a green backpack. Several participants mentioned the use of color, and participants tied this to contributing to a more sophisticated design. One participant noted, “The mountain shape is a more advanced design compared to the traditional progress bar. The mountain shape has different levels color coded as well as the review questions identified by receiving the shield showing the user has mastered the topic.” Another user tied the use of color directly to feelings of excitement by stating, “The gaining shields and different colors and climbing the mountain kind of makes me

excited to advance through the course.” These indicators support the design principle that multiple colors can be eye-catching and motivating.

Design Principle: Progress bars utilizing multiple colors can be eye-catching and motivating.

To instantiate this design principle, designers were strategic in their color choice, placing colors that were distinct from each other in adjacent places in the design. One participant noted the appropriate use of color combinations to support users with color blindness by stating, “I do have slight color blindness, so I can't tell shades apart. So you've done a good job separating [colors].” However, another user noted having a slight issue where the colors transition from green to blue. These statements demonstrate the need for designers to consider color combinations in progress bar design, and the need for designers to include color combinations that meet the needs of users with color blindness or low vision.

Design Principle: Progress bars should include color combinations that meet the needs of users with color blindness or low vision.

When discussing color concerns, one of the participants expressed favoring the color coding to indicate progression through different sections of the module; however, the participant did note concerns about the amount of different colors used. The participant stated, “I liked that with each milestone, you meant that the color changed, and I thought it was just a bit more cohesive. But I agree about, I think, with the mountain in the background, while I'm also trying to look at the steps, just all of the different colors in that aspect was distracting a bit.” This observation supports the need for designers to be strategic in their use of color when making choices about progress bar design.

Design Principle: Progress bar colors should be bright enough to be easy to see and distinct from other content but not so bright that they are distracting.

Aside from motivation, users commented the most on the theming of Version B. The module housing the progress bar was themed as a jungle adventure, with learning visuals using exotic animals. This theme was carried over to progress bar Version B through the mountain shape and a characterized hiker. Since the learning content dealt with steganography, a topic associated with cybersecurity, the hiker collected shields along the progress bar when learners completed a learning check. Participants noted and commented on this connection between the progress bar design and the content. One participant noted, “I think the mountain fits in better

with the adventure than a standard progress bar.” Another user stated, “The mountain shape is more appealing to the human eye and provided an increase in motivation because of the different levels and achievements along the way.” Another participant noted that the theming generated more interest, “Progress Bar B was more interesting as it had much more components to completing the module with the concept of the man climbing the mountain, rather than just a green line slowly inching across the screen. There was more of a story and rewards with the mountain progress bar.” The use of characterization was noted by another participant who felt that the theming allowed for more of an emotional investment from users. “The mountain progress bar gives a sense of personhood to the progress. The man climbing the mountain provides a story for the user to follow along with and allows the user to attach emotion to the progress. In contrast, the traditional progress bar provides no more emotion than one would expect of a rectangle on a screen.” These comments demonstrate the need for progress bar design to complement and connect with the rest of the content.

Design Principle: Progress bar design should complement and connect with the rest of the content.

Ultimately, the progress bar in a learning module is intended to indicate user progress. Therefore, there is a need for the user to be able to gauge progress easily. Four users noted that the green bar was easier to understand at a glance and suggested that the milestones in Version B be more linear instead of zigzagging up the mountain. No users reported an issue understanding Version B. In fact, one felt Version B provided clearer expectations and stated, “The color path with checkpoints is like a reward system and lowers anxiety with clear expectations. The shields with completion of questions is also exciting.” This feedback reiterates the need for progress bars to be easy to understand.

Design Principle: Progress bars should be easy to understand.

Both progress bars were placed in the lower left corner of the screen. Version B was scaled to cover about 2% of the screen. The participant who noted that the use of color was distracting did tie part of their concern to the size of the progress bar. The participant stated, “Part of me kind of wants it to be larger. I do still stand by the fact that it's distracting, but I feel like it would be somehow less distracting if it was larger because I could see it like, easier, I guess, and then I can focus on the rest of the slides.” This statement supports the principle that

designers must be conscientious of the size of the progress bar when integrating the bar into learning modules.

Design Principle: Progress bar size should complement the design of the progress bar and the rest of the content.

Another principle tied to the size of the progress bar is the need for it to scale in size, as screen size will vary from device to device. Since participants were not asked to view the designs across multiple devices, the scalability of the bar was not addressed by users, except regarding the previous comment about the design needing to take up more of a percentage on the screen. However, one user did complement the complexity of the design by stating, “Progress Bar B was more interesting, as it had much more components to completing the module with the concept of the man climbing the mountain, rather than just a green line slowly inching across the screen. There was more of a story and rewards with the mountain progress bar.” This comment, along with the comment about the percentage of screen real estate being used by the progress bar, supports the principle that complexity and scalability should be a consideration in progress bar design.

Design Principle: Progress bar complexity should scale well across varying screen sizes and display formats.

Version B used stepping stones that colored in as the user progressed. These were color coded to indicate segments of the learning module and updated on a frame by frame basis, meaning the transparency of the stones decreased (became more solid) when the user completed a section. This indication of progression was complimented by participants. One participant noted, “I think with the mile markers, it kind of gives an incentive to keep going where [the traditional] does not do that.” Another participant also stated, “So it actually feels like a game. It’s like, oh, I finished that part. Now I’m onto a new part.” The user feedback about the frequent updating of the progress bar supports the design principle that progress bars should update frame by frame to support users’ knowledge of their progress.

Design Principle: Progress bars should update frame by frame to better reflect user progression.

In addition to updating frequently, users also appreciated the bar’s indication of the remaining content to be covered. One participant noted, “I do like how the mountain shows where the learning checks will be, the traditional progress bar only shows how far away you are

from the finish.” Another participant stated, “The mountain shape gives the user something to look forward to as they climb the mountain and advance throughout the course. It’s also a more visual representation of the levels.” This preference for a visual representation of levels supports the design principle that progress bars should be segmented to accurately reflect the amount of content remaining.

Design Principle: Progress bar segmentation should be representative of the content remaining in the task.

Following the sessions, overall findings were shared with the participants for member checking. Of the eleven participants, eight responded to invitations to review the findings. All eight respondents reported that the findings accurately reflected their experience, with the only suggested change being a rewording of one of the figure headings for clarity. This suggestion was accepted by the researchers.

CHAPTER 7

DISCUSSION

The objectives of the research were to fill a gap related to the intentional design of progress bars, especially in the areas of education and training, by developing design principles to assist designers and developers in producing progress bars to improve motivational mechanics in online educational settings. Specifically, the research aimed to answer the following research question: *What progress bar design principles improve motivational mechanics?*

To support these aims, the researchers leveraged a theoretical foundation built on the relationship among motivation, motivational affordance, and UX and identified design elements tied to information visualization in order to develop principles for progress bar design. The resulting design elements took into account the pragmatic and hedonic qualities of progress bars, as well as factors such as color psychology, cognitive cueing, and accessibility guidelines.

The resultant drafts of the principles were refined through two rounds of feedback using instantiations of design principles in progress bar prototypes. Round 1 feedback was collected from curriculum developers, educators, and UX designers. Round 2 feedback was collected from university students. The resultant feedback was further incorporated in the design principles, with these final design principles being incorporated into another instantiation for evaluation.

The final evaluation demonstrated how incorporating the design principles could improve motivational mechanics, with the instantiation of the design principles performing better on average than the traditional progress bar on the UEQ-S survey overall, being rated higher than the traditional progress bar on average by user. Additionally, the instantiation performed several points higher on average than the traditional progress bar in the areas of exciting, interesting, inventive, and leading edge. These findings were supported by open-ended participant feedback, where participants described the instantiation of the design principles as being new, exciting, immersive, and game-like. Table 19 lists the evaluated principles.

Table 19. *Design Principles for Progress Bars to Improve Motivational Mechanics*

Final Design Principles
Progress bars utilizing multiple colors can be eye catching and motivating.
Progress bars should include color combinations that meet the needs of users with color blindness or low vision.
Progress bar colors should be bright enough to be easy to see and distinct from other content but not so bright that they are distracting.
Progress bar design should complement and connect with the rest of the content.
Progress bar text, if used, should have a mix of uppercase and lowercase text.
Progress bars should be easy to understand.
Progress bar size should complement the design of the progress bar and the rest of the content.
Progress bar complexity should scale well across varying screen sizes and display formats.
Progress bars should update frame by frame to better reflect user progression.
Progress bar segmentation should be representative of the content remaining in the task.

To date, researchers have not provided competing design principles for progress bars to improve motivational mechanics in an online educational environment. However, researchers have provided design principles for other gaming elements, such as leaderboards, that were incorporated into online learning (Bovee, 2022). In the case of leaderboards, the design principles dealt with clear instructions, challenge skill balance, and timely feedback. The provided design principles for progress bars align with the principles in that they both address the need to for user feedback. However, due to the differing nature of the two elements, the leaderboard design principles focus on user engagement (goal orientation and social connectivity), while the design principles for progress bars lend themselves to promoting user motivation tied to challenge, interactivity, achievement, and reinforcement (Nah et al., 2019).

As noted above, Nah et al. (2019) provided initial guidance on progress bars' alignment with the principles of gamification, having determined the tool's role in gamifying learning environments. However, Nah et al. (2019) did not provide any recommendations on how to leverage progress bar design to activate or improve any of the aligned principles. Sailer et al. (2017) also discussed the role of a progress bar in gamified environments, but they only explore the role of the progress bar as a feedback device and its contribution to game-like user experiences, without providing guidance on the selection, application, implementation, or integration of the design element.

Spence (2014) also provided guidance on information visualization, noting the need for visuals to provide representation, presentation, and interaction; however, the author focuses on identifying these as components in the information visualization process and does not offer specific principles for exemplifying these components in progress bars.

As far as new contributions to research and practice, this research can be classified in the improvement domain of the Design Science Research Knowledge Contribution Framework, as that domain is focused on providing a new solution for a known problem (Gregor & Hevner, 2013). The improvement domain best aligns, as it targets creating “better solutions in the form of more efficient and effective products, processes, services, technologies, or ideas” (Gregor & Hevner, 2013). It also answers the call for researchers to “redesign systems to motivate new behaviors in a consistent, generalizable, ethical, and theoretically justifiable way” (Landers, 2019). Based on theory tied to gamification and information visualization, the development of the principles demonstrate how information visualization research can be used to impact motivational affordance to improve UX. The design principles create an improvement in the quality of progress bars in three distinct areas:

1. Progress bar representation: Principles targeting the color, shape, and text of progress bars provide designers and researchers with research-based guidance on how those characteristics should be represented to indicate user progress to better motivate users, such as in online learning modules.
2. Progress bar presentation: Principles targeting the placement and size of progress bars offer direction for integrating the tool into online environments to better motivate users.
3. Progress bar interaction: Principles targeting the interactive capabilities of the progress bars offer designers and researchers strategies to leverage animation to better motivate users in online environments.

These three areas of progress bar design support increasing user motivation through the selection, application, implementation, and/or integration of progress bars as design elements in online environments, including educational environments. It leverages a theory-based approach, addressing both the pragmatic and hedonic qualities of UX (Schrepp et al., 2017). These principles are new findings and provide a significant contribution to gamification research, which support the gamification of online environments, including educational environments.

CHAPTER 8

CONCLUSION

This research fills a gap in the gamification of online modules by providing design principles of progress bars to improve motivational mechanics. The study builds on utilizing characteristics of information visualization to positively impact motivational mechanics, specifically, artifactual motivational affordance (Deterding, 2011; Spence, 2014). Initial design principles were derived from streams of literature and refined through the use of an instantiation to gather user feedback. Feedback was collected in two rounds: with one round coming from experts in the fields of education, curriculum design, and UX and the other round coming from adult users. This feedback was used to further refine the design principles, with the final design principles being integrated into an instantiation for summative evaluation by a user-based focus group. The evaluation supported the design principles, as the instantiation of the design principles performed better overall than the traditional progress bar in the UEQ-S survey, which measures the six aspects of UX, and focus group interviews.

The design principles support the representation, presentation, and interaction of progress bars to improve motivational mechanics by increasing the appeal of progress bar itself. This appeal was measured by aligning the instantiation's performance with indicators of UX to ensure that the design principles accounted for goal attainment (pragmatic quality) and fun (hedonic quality).

There are several limitations that must be recognized in order to frame the findings presented. The first limitation is that the results were derived from a series of focus groups utilizing just one instantiation of the design principles, with the final evaluation limited to college student participation. This limits the generalizability of the findings. Additionally, the design principles were instantiated into a cybersecurity-based learning module and could have resulted in different outcomes based on the content and other components tied to the learning module. Finally, the research did not capture the usability of the design principles by other researchers.

Future research is needed across a variety of audiences and content areas to better inform on the usefulness of the design principles. Instantiations were only tested with adult learners. It

would be beneficial to determine how using the design elements can impact other audiences. Tied to this is the benefit of broadening the research to increase the number of participants and other research approaches, including quantitative methods. Further, future researchers could build on the findings by expanding the research to include larger audiences and different evaluation methods. Another area of future research is the leveraging of the design principles in the context of other content areas and other learning module formats and designs. This research would help understand the generalizability of the findings and determine how the design principles work when applied to different content and screen formats. Future research into how well other researchers can interpret and utilize the design principles to improve motivation would also be beneficial. The use of text, addressed in the design principles, was not tested and would also be an area of future study.

In conclusion, this research provides insight as to how to improve motivational mechanics in online modules through progress bar design. Building on gamification and information visualization theories, as well as feedback from experts and users, the design principles offer an approach for progress bar development to support the use of the element for improving motivation as evidenced through an improved user experience. The design principles will assist researchers and developers in being intentional in their inclusion of progress bars as a gamification element by addressing design needs in the areas of representation, presentation, and interaction. The design principles offer the opportunity for better design of a gamification elements that is tied to challenge, interactivity, achievement, and reinforcement, qualities that are desired across a variety of environments, benefitting future gamification research as well as designers wishing to leverage gamification in online settings, such as education.

This DSRM approach involved two rounds of feedback as well as a focus group evaluation. The end result was 10 design principles for progress bars targeting increasing motivational mechanics.

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APPENDICES

APPENDIX A: INTERVIEW QUESTIONS

1. Which of the progress bar designs did you find most supportive of the user experience? Why?
2. Which of the progress bar designs did you find easier to understand? Why?
3. Which of the progress bar designs did you find the more efficient? Why?
4. Which of the progress bar designs did you find clearer? Why?
5. Which of the progress bar designs did you find the more exciting? Why?
6. Which of the progress bar designs did you find the more interesting? Why?
7. Which of the progress bar designs did you find the more inventive? Why?
8. Which of the progress bar designs did you find the more leading edge? Why?
9. Did Progress Bar A increase your motivation to complete the task? What about it promoted/undermined motivation?
10. Did Progress Bar B increase your motivation to complete the task? What about it promoted/undermined motivation?
11. Is there anything else about the designs you would like to let me know?